GRUNDY DOMINATION OF KNESER GRAPHS AND GRAPH PRODUCTS

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Abstract. The Grundy domination number of a graph is at most the minimum rank of its closed neighbourhood pattern over any field, and for a strong product at most the product of the minimum ranks of the factors. For Kneser graphs a matrix over a prime field gives $\gamma_{\mathrm{gr}}(K(n,r)) = \binom{n-1}{r-1}$ whenever a prime power at least r divides $n-2r$, which proves the conjecture of Campêlo and Severín for all but finitely many n for each r; five further small values are determined. For products we prove $\gamma_{\mathrm{gr}}(C_k \boxtimes C_\ell) = (k-2)(\ell-2)$ and the analogous formulas for many strong and direct products. Both product conjectures of Brešar et al. are false, and the largest ratio is $n^{1/2+o(1)}$ in the order n of the product. Incidence graphs of projective planes show that bipartite graphs without 4-cycles can have Grundy total domination number $o(n)$.

1. Introduction

1.1. **Dominating sequences.** All graphs are finite and simple. For a vertex v of a graph G we write $N(v)$ for its open and $N[v] = N(v) \cup \{v\}$ for its closed neighbourhood. A sequence $(v_1, \dots, v_k)$ of vertices is a *dominating sequence* if

$$N[v_i] \setminus \bigcup_{j<i} N[v_j] \neq \emptyset \qquad (1 \le i \le k), \tag{1.1}$$

and a *total dominating sequence* if the same holds with open neighbourhoods,

$$N(v_i) \setminus \bigcup_{j<i} N(v_j) \neq \emptyset \qquad (1 \le i \le k). \tag{1.2}$$

A vertex in the set on the left is a *footprint* of v_i.[1] The *Grundy domination number* $\gamma_{\mathrm{gr}}(G)$ is the largest length of a dominating sequence, and the *Grundy total domination number* $\gamma^t_{\mathrm{gr}}(G)$ is the largest length of a total dominating sequence; the second is defined for graphs without isolated vertices. These invariants were introduced by Brešar, Gologranc, Milanič, Rall and Rizzi [10] and by Brešar, Henning and Rall [11]. A longest sequence of either kind is a dominating, respectively total dominating, set, since an undominated vertex could be appended. So $\gamma_{\mathrm{gr}}(G)$ and $\gamma^t_{\mathrm{gr}}(G)$ are the largest sizes of such sets that can be built greedily while every new vertex enlarges the dominated set; they are the upper ends of the scale on which the domination number $\gamma(G)$ is the lower end. Computing either invariant is NP-hard [10, 11].

The two invariants are tied to zero forcing and to minimum rank. Brešar, Bujtás, Gologranc, Klavžar, Košmrlj, Patkós, Tuza and Vizer [9] showed that the zero forcing number of G is $|V(G)|$ minus a Grundy-type invariant of the same family, and Lin [18] showed that each of the Grundy invariants is bounded above by the minimum rank of a family of real symmetric matrices attached to the graph. The present paper turns this upper bound into a tool for graph products.

2020 *Mathematics Subject Classification.* Primary 05C69, 05C76; Secondary 05C50, 05D05, 15A03, 51E15.

Key words and phrases. Grundy domination; Kneser graphs; graph products; minimum rank.

[1] The vertices of either kind of sequence are distinct, because a repeated vertex has no footprint. A vertex may have several footprints, and any one of them certifies the step.

1.2. **Products and the two conjectures.** In the strong product $G \boxtimes H$, two distinct vertices (g,h) and (g',h') are adjacent when $g' \in N[g]$ and $h' \in N[h]$, so that

$$N_{G\boxtimes H}[(g,h)] = N_G[g] \times N_H[h].$$

In the direct product $G \times H$ they are adjacent when $g' \in N(g)$ and $h' \in N(h)$, so that $N_{G\times H}((g,h)) = N_G(g) \times N_H(h)$. Listing the pairs of two longest sequences of the factors in lexicographic order gives

$$\gamma_{\rm gr}(G \boxtimes H) \ge \gamma_{\rm gr}(G)\,\gamma_{\rm gr}(H), \qquad \gamma^t_{\rm gr}(G \times H) \ge \gamma^t_{\rm gr}(G)\,\gamma^t_{\rm gr}(H), \tag{1.3}$$

for all graphs G and H (Lemma 2.3). Brešar et al. conjectured that both inequalities are equalities.

Conjecture 1.1 ([8, Conjecture 22])**.** *For all graphs G and H, $\gamma_{\rm gr}(G \boxtimes H) = \gamma_{\rm gr}(G)\,\gamma_{\rm gr}(H)$.*

Conjecture 1.2 ([7, Conjecture 2.2])**.** *For all graphs G and H without isolated vertices, $\gamma^t_{\rm gr}(G\times H) = \gamma^t_{\rm gr}(G)\,\gamma^t_{\rm gr}(H)$.*

Conjecture 1.1 was proved when one factor is a caterpillar [8] and, by Bell, Driscoll, Krop and Wolff [3], when one factor is a forest.[2] The exact value of $\gamma_{\rm gr}(C_k \boxtimes C_\ell)$, the Grundy domination number of the strong torus, was left open in [8], where the bound $\gamma_{\rm gr}(C_k \boxtimes C_\ell) \le (k-2)(\ell-1)$ was proved and the value $(k-2)(\ell-2)$ predicted by Conjecture 1.1 was stated as conjectural. Conjecture 1.2 was proved in [7] when one factor is a tree, when both factors are cycles, and in a few other cases, and the authors noted that it is equivalent to Conjecture 1.1 through a hypergraph formulation.

1.3. **Results.** For a field $\mathbb{F}$ and a graph G let $\mathrm{mr}_{\mathbb{F}}(G)$ be the least rank of a square matrix M over $\mathbb{F}$, with rows and columns indexed by $V(G)$, whose nonzero entries are exactly the pairs (u,v) with $v \in N[u]$; and let $\mathrm{mr}^\circ_{\mathbb{F}}(G)$ be the least rank of such a matrix whose nonzero entries are exactly the pairs with $v \in N(u)$. The diagonal of the first kind of matrix is nonzero and that of the second is zero. Unlike the minimum rank of the theory of zero forcing, the matrices need not be symmetric, the field is arbitrary, and the diagonal is prescribed.

Theorem A (rank bound)**.** *For every field $\mathbb{F}$ and all graphs G and H,*

$$\gamma_{\rm gr}(G \boxtimes H) \le \mathrm{mr}_{\mathbb{F}}(G)\ \mathrm{mr}_{\mathbb{F}}(H), \qquad \gamma^t_{\rm gr}(G \times H) \le \mathrm{mr}^\circ_{\mathbb{F}}(G)\ \mathrm{mr}^\circ_{\mathbb{F}}(H).$$

Theorem A rests on the fact that a dominating sequence produces a triangular submatrix with nonzero diagonal (Lemma 2.1), and that the Kronecker product of two matrices of the right shape has the shape of the product graph. It singles out the graphs for which the bound is attained. Call G *$\mathbb{F}$-tight* if $\gamma_{\rm gr}(G) = \mathrm{mr}_{\mathbb{F}}(G)$, and *totally $\mathbb{F}$-tight* if $\gamma^t_{\rm gr}(G) = \mathrm{mr}^\circ_{\mathbb{F}}(G)$. Strong products of $\mathbb{F}$-tight graphs are $\mathbb{F}$-tight and satisfy Conjecture 1.1, and direct products of totally $\mathbb{F}$-tight graphs satisfy Conjecture 1.2 (Corollary 2.5). Section 3 finds many tight graphs. The cycles are among them, which settles the torus.

Theorem B (tori and Kneser graphs)**.** *Let $r \ge 1$ and $k_1, \dots, k_r \ge 3$. Then*

$$\gamma_{\rm gr}(C_{k_1} \boxtimes C_{k_2} \boxtimes \cdots \boxtimes C_{k_r}) = \prod_{i=1}^{r}(k_i - 2).$$

More generally, the Grundy domination number of a strong product of cycles, paths, complete graphs and complete bipartite graphs is the product of the Grundy domination numbers of the factors. For every $n \ge 6$ the Kneser graph $K(n,2)$ is tight over $\mathbb{F}_2$ if n is even and over $\mathbb{F}_p$ for every odd prime p dividing $n-4$ if n is odd; so, for instance, $\gamma_{\rm gr}(K(n,2) \boxtimes C_k) = (n-1)(k-2)$

[2] A proof for all graphs was posted as arXiv:2212.04565 and withdrawn by its authors in its second version.

whenever n is even and $k \geq 3$ is odd, and $\gamma_{\mathrm{gr}}(K(n,2) \boxtimes K(m,2)) = (n-1)(m-1)$ whenever $n, m \geq 6$ are even.

For Kneser graphs of higher rank the same method needs a different matrix, built from the binomial coefficients $\binom{|S\cap T|-1}{r-1}$. Write $\operatorname{lcm}(1,\dots,r-1)$ for the least common multiple.

Theorem C (Kneser graphs)**.** *Let $r \geq 2$ and $n \geq 2r+1$. If some prime power $p^a \geq r$ divides $n-2r$, equivalently if $n-2r$ does not divide $\operatorname{lcm}(1,\dots,r-1)$, then $K(n,r)$ is $\mathbb{F}_p$-tight and*

$$\gamma_{\mathrm{gr}}(K(n,r)) = \alpha(K(n,r)) = \binom{n-1}{r-1}.$$

So the conjecture of Campêlo and Severín [14] *that $\gamma_{\mathrm{gr}}(K(n,r)) = \binom{n-1}{r-1}$ for all $n \geq 3r$ holds for every $n > 2r + \operatorname{lcm}(1,\dots,r-1)$, for every $n \geq 3r$ when $r \leq 3$, and for every $n \geq 3r$ except $n = 14$ when $r = 4$. Moreover $\gamma_{\mathrm{gr}}(K(8,3)) = 22$, $\gamma_{\mathrm{gr}}(K(9,4)) = 77$, $\gamma_{\mathrm{gr}}(K(10,4)) = 93$, $\gamma_{\mathrm{gr}}(K(11,4)) = 121$ and $\gamma_{\mathrm{gr}}(K(11,5)) = 296$.*

Theorem C bounds the unspecified threshold of Brešar, Kos and Torres [13] by $2r + \operatorname{lcm}(1,\dots,r-1) + 1$, and the five exact values show that the dominating sequences found by tabu search in [14] are optimal. Strong products of Kneser graphs that are tight over the same prime field satisfy Conjecture 1.1 (Corollary 5.3). A version of the rank bound with a free diagonal also gives $\gamma^{Z}_{\mathrm{gr}}(K(6,2)) = 5$ for the Z-Grundy number, so the zero forcing number of $K(6,2)$ is 10 (Proposition 4.12).

The value $\gamma_{\mathrm{gr}}(K(n,2)) = n-1$ for $n \geq 6$ is due to Brešar, Kos and Torres [13]; the rank argument of Proposition 3.8 gives a short proof of the upper bound. The direct-product analogue holds for totally tight graphs, which include all cycles, trees, complete graphs and complete bipartite graphs (Theorem 7.4); it extends the theorem of [7] for two cycles to any number of cycles, trees and complete graphs.

Tightness is not universal, and the failure of tightness is exactly what the conjectures miss. The next theorem shows that both conjectures fail and that no constant multiple of the predicted value is an upper bound.

Theorem D (the conjectures fail)**.**

(i) *For every $C > 0$ there are graphs G, H with $\gamma_{\mathrm{gr}}(G \boxtimes H) > C\,\gamma_{\mathrm{gr}}(G)\,\gamma_{\mathrm{gr}}(H)$ and graphs G', H' without isolated vertices with $\gamma^{t}_{\mathrm{gr}}(G' \times H') > C\,\gamma^{t}_{\mathrm{gr}}(G')\,\gamma^{t}_{\mathrm{gr}}(H')$.*

(ii) *Let H_N be the graph on $\{0,1,\dots,N-1\}$ in which u and v are adjacent when $|u-v| \equiv 1 \pmod 3$. For $N \geq 4$, $\gamma_{\mathrm{gr}}(H_N) = \lfloor N/3 \rfloor + 2$, while $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N) \geq N$; so Conjecture 1.1 fails for the octahedron $K_{2,2,2}$ and H_N when $N \in \{11, 13, 14\}$ or $N \geq 15$.*

(iii) *Measured against the order n of the product, the largest ratio is $n^{1/2+o(1)}$ for both products: $\gamma_{\mathrm{gr}}(G \boxtimes H)/(\gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H)) \leq n^{1/2}$ for all G, H with $|V(G)||V(H)| \leq n$, and for every $\varepsilon > 0$ the ratio exceeds $c_\varepsilon n^{1/2-\varepsilon}$ for suitable such G, H; the same holds for γ^{t}_{gr} and direct products.*

The octahedron is $\mathbb{F}$-tight over every field with at least five elements (Proposition 3.5). So in part (ii) the graph H_N cannot be tight over any field, and indeed $\gamma_{\mathrm{gr}}(H_N) < \mathrm{mr}_{\mathbb{F}}(H_N)$ for every $\mathbb{F}$ and every N in the range of (ii) (Corollary 6.11). For the direct product, the passage from closed to open neighbourhoods is a doubling construction: if $\mathsf{B}(G)$ is the bipartite graph on $V(G) \times \{0,1\}$ in which $(u,0)$ is adjacent to $(v,1)$ when $v \in N[u]$, then $\gamma^{t}_{\mathrm{gr}}(\mathsf{B}(G)) = 2\gamma_{\mathrm{gr}}(G)$ and $\mathsf{B}(G) \times \mathsf{B}(H)$ is two disjoint copies of $\mathsf{B}(G \boxtimes H)$ (Proposition 7.5). The smallest counterexample to Conjecture 1.1 that we know has 66 vertices, and the smallest one to Conjecture 1.2 has 264.

The last theorem concerns a question of Brešar, Henning and Rall [11, Question 1]. They showed that there is no constant $c > 0$ with $\gamma^{t}_{\mathrm{gr}}(G) \geq c|V(G)|$ for all bipartite graphs without

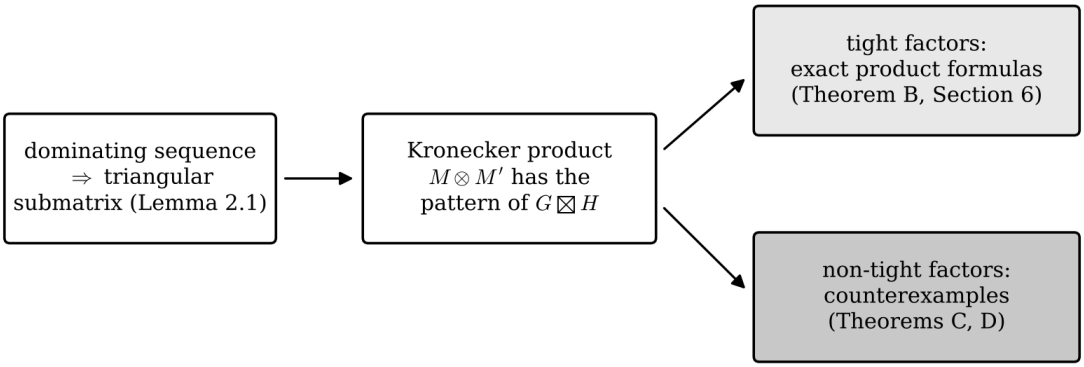


FIGURE 1. The structure of the paper. A dominating sequence gives a triangular submatrix (left), the Kronecker product carries matrices to products (centre), and tight factors give exact product formulas, while non-tight factors give the counterexamples (right).

open twins, noticed that every vertex of their examples lies on a 4-cycle, and asked whether such a constant exists for bipartite graphs without 4-cycles and with minimum degree at least 2.

Theorem E (projective planes)**.** *Let $q = p^e$ be a prime power and let* $\mathrm{Inc}(q)$ *be the point–line incidence graph of the projective plane* $\mathrm{PG}(2, q)$*. It is bipartite,* $(q+1)$*-regular, has girth* 6 *and* $n = 2(q^2 + q + 1)$ *vertices, and*

$$\gamma_{\mathrm{gr}}^{t}(\mathrm{Inc}(q)) \le 2\Big(\tbinom{p+1}{2}^{e} + 1\Big).$$

In particular $\gamma_{\mathrm{gr}}^{t}(\mathrm{Inc}(2^e)) \le 2(3^e + 1)$*, and* $\gamma_{\mathrm{gr}}^{t}(\mathrm{Inc}(p^e))/|V(\mathrm{Inc}(p^e))| \to 0$ *as* $e \to \infty$ *for every fixed prime* p*. So the answer to the question of Brešar, Henning and Rall is negative, even for regular graphs of girth* 6 *and arbitrarily large degree. For* $q \in \{2, 3, 4, 8\}$ *the bound is attained.*

1.4. **How the paper is organised.** Section 2 proves the rank lemma and Theorem A. Section 3 constructs tight graphs: complete, complete bipartite and cocktail party graphs, cycles, paths and Kneser graphs $K(n, 2)$. Section 4 proves Theorem C for Kneser graphs of every rank, determines five small Kneser graphs and treats the Z-Grundy number, and Section 5 assembles the tight graphs into Theorem B and a table of exact values. Section 6 proves the strong-product part of Theorem D, first with two graphs on the five-element subsets of an m-set, whose factor bounds come from incidence matrices over two different fields, then shows that a choice of two larger primes gives the optimal order of magnitude, and finally treats the octahedron and the graphs H_N. Section 7 treats direct products and Grundy total domination and proves the remaining part of Theorem D. Section 8 proves Theorem E. Section 9 reports exact computations and a search for small counterexamples, and Section 10 lists the open problems, with a table of the status of every conjecture and question treated here (Table 6). The appendices contain explicit sequences and matrices and describe the independent machine checks.

The method is linear algebra over finite fields together with elementary counting. Counting arguments attached to a finite structure or to a group action appear also in the author's earlier work [4, 5, 6].

2. Triangular submatrices and the rank bound

2.1. **Pattern matrices.** Let G be a graph and $\mathbb{F}$ a field. A square matrix M over $\mathbb{F}$ with rows and columns indexed by $V(G)$ is a *closed pattern matrix* of G if

$$M_{uv} \neq 0 \iff v \in N[u] \qquad (u, v \in V(G)),$$

and an *open pattern matrix* of G if $M_{uv} \neq 0 \iff v \in N(u)$. The two sets of matrices are denoted $\mathcal{S}_{\mathbb{F}}[G]$ and $\mathcal{S}_{\mathbb{F}}(G)$, and

$$\mathrm{mr}_{\mathbb{F}}(G) = \min\{\operatorname{rank} M : M \in \mathcal{S}_{\mathbb{F}}[G]\}, \qquad \mathrm{mr}^{\circ}_{\mathbb{F}}(G) = \min\{\operatorname{rank} M : M \in \mathcal{S}_{\mathbb{F}}(G)\}.$$

The matrices are not required to be symmetric. If G has no isolated vertex, then $\mathcal{S}_{\mathbb{F}}(G)$ is nonempty; $\mathcal{S}_{\mathbb{F}}[G]$ always contains $A + I$, where A is the adjacency matrix, and the identity of $\mathbb{F}$ is used for the nonzero entries.

Two elementary remarks will be used repeatedly. First, if $\mathbb{F} \subseteq \mathbb{F}'$ is a field extension, then a matrix over $\mathbb{F}$ has the same rank over $\mathbb{F}'$, so

$$\mathrm{mr}_{\mathbb{F}'}(G) \leq \mathrm{mr}_{\mathbb{F}}(G), \qquad \mathrm{mr}^{\circ}_{\mathbb{F}'}(G) \leq \mathrm{mr}^{\circ}_{\mathbb{F}}(G). \tag{2.1}$$

Second, both invariants are additive over disjoint unions, by block-diagonal matrices, and the Grundy invariants are additive as well, because a sequence in a disjoint union is legal exactly when its restriction to each component is legal.[3]

2.2. **The rank lemma.**

Lemma 2.1. *Let G be a graph and $\mathbb{F}$ a field. Then $\gamma_{\mathrm{gr}}(G) \leq \mathrm{mr}_{\mathbb{F}}(G)$, and if G has no isolated vertex, $\gamma^{t}_{\mathrm{gr}}(G) \leq \mathrm{mr}^{\circ}_{\mathbb{F}}(G)$.*

Proof. Let $(v_1, \ldots, v_k)$ be a dominating sequence and choose a footprint w_i of v_i for each i. If $j < i$, then $w_i \notin N[v_j]$, while $w_j \in N[v_j]$; hence $w_i \neq w_j$, and the footprints are distinct. Let $M \in \mathcal{S}_{\mathbb{F}}[G]$ and consider the $k \times k$ submatrix T with rows $v_1, \ldots, v_k$ and columns $w_1, \ldots, w_k$, in this order. Its diagonal entries $M_{v_i w_i}$ are nonzero since $w_i \in N[v_i]$, and its entries above the diagonal vanish, because for $j < i$ the entry in row v_j and column w_i is $M_{v_j w_i} = 0$ as $w_i \notin N[v_j]$. So T is lower triangular with nonzero diagonal, $\operatorname{rank} T = k$, and $k \leq \operatorname{rank} M$. Taking a longest sequence and a matrix of least rank gives $\gamma_{\mathrm{gr}}(G) \leq \mathrm{mr}_{\mathbb{F}}(G)$. The total case is identical, with open neighbourhoods and $M \in \mathcal{S}_{\mathbb{F}}(G)$. □

For real symmetric matrices, and with the diagonal free in the appropriate way, the inequality is Lin's [18, Theorem 3.1]. The argument needs neither symmetry nor any particular field, but it does need the nonzero entries to be exactly the prescribed ones: a footprint w_i is in general a neighbour of v_i and not v_i itself, so the entries M_{uv} with $v \in N(u)$ must be nonzero, and the entries outside the neighbourhoods must vanish (Figure 2).

Remark 2.2. The argument also shows what $\gamma_{\mathrm{gr}}(G)$ is in matrix terms: it is the largest k for which there are distinct vertices $v_1, \ldots, v_k$ and distinct vertices $w_1, \ldots, w_k$ such that $w_i \in N[v_i]$ for all i and $w_i \notin N[v_j]$ for $j < i$. In other words, $\gamma_{\mathrm{gr}}(G)$ is the size of the largest lower triangular submatrix with nonzero diagonal, after independent permutations of rows and columns, of the 0/1 matrix $A + I$. The rank bound replaces this combinatorial quantity, which is not monotone under taking matrices with the same pattern, by a linear-algebraic one that is.

[3] The domination steps in different components do not interact: $N[v]$ lies in the component of v.

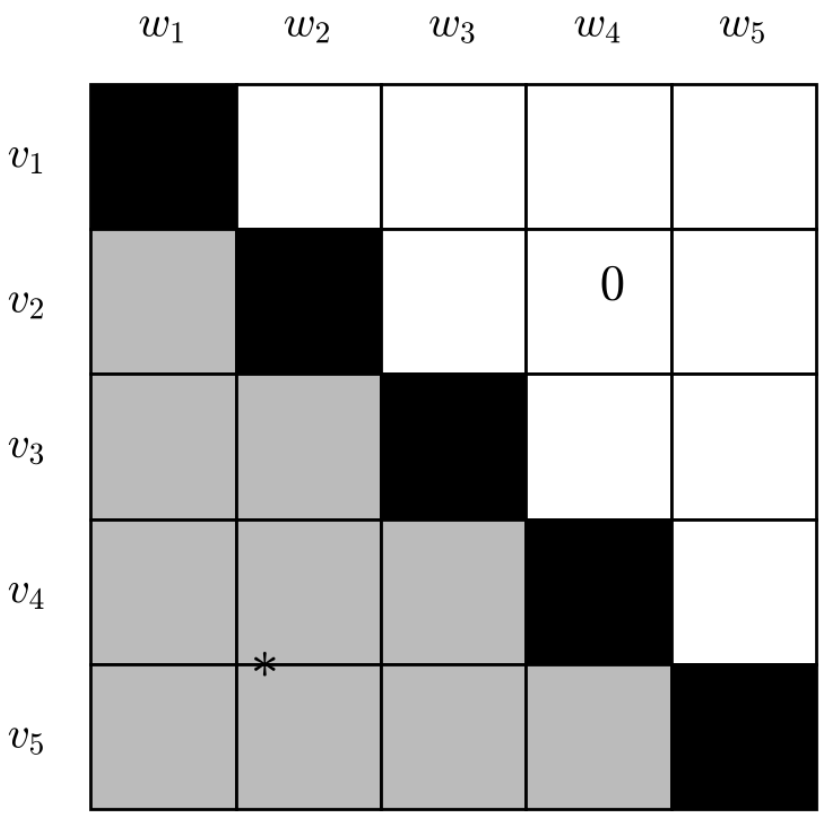


FIGURE 2. A dominating sequence $(v_1, \ldots, v_5)$ with footprints $w_1, \ldots, w_5$ selects a lower triangular submatrix with nonzero diagonal from any pattern matrix. The black cells are the diagonal, the white cells above it vanish, and the grey cells below it are unconstrained.

2.3. **Lower bounds for products.**

Lemma 2.3. *For all graphs G and H, $\gamma_{\mathrm{gr}}(G \boxtimes H) \geq \gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H)$; if G and H have no isolated vertices, $\gamma^t_{\mathrm{gr}}(G \times H) \geq \gamma^t_{\mathrm{gr}}(G)\gamma^t_{\mathrm{gr}}(H)$.*

Proof. Let $(g_1, \ldots, g_a)$ and $(h_1, \ldots, h_b)$ be dominating sequences of G and H with footprints x_i and y_j. List the pairs (g_i, h_j) in lexicographic order. The pair (x_i, y_j) lies in $N[g_i] \times N[h_j] = N[(g_i, h_j)]$. An earlier pair (g_s, h_t) has either $s < i$, and then $x_i \notin N[g_s]$, or $s = i$ and $t < j$, and then $y_j \notin N[h_t]$; in both cases $(x_i, y_j) \notin N[(g_s, h_t)]$. So (x_i, y_j) is a footprint of (g_i, h_j) and the sequence has length ab. The total case is the same with open neighbourhoods and the direct product. □

The first inequality is [8, Proposition 21] and the second is [7, Lemma 2.1]. The same proof works for any number of factors: for graphs $G_1, \ldots, G_r$,

$$\gamma_{\mathrm{gr}}(G_1 \boxtimes \cdots \boxtimes G_r) \geq \prod_{i=1}^{r} \gamma_{\mathrm{gr}}(G_i), \qquad \gamma^t_{\mathrm{gr}}(G_1 \times \cdots \times G_r) \geq \prod_{i=1}^{r} \gamma^t_{\mathrm{gr}}(G_i). \tag{2.2}$$

2.4. **The rank bound for products.**

Proof of Theorem A. Let $M \in \mathcal{S}_{\mathbb{F}}[G]$ and $M' \in \mathcal{S}_{\mathbb{F}}[H]$ have ranks $\mathrm{mr}_{\mathbb{F}}(G)$ and $\mathrm{mr}_{\mathbb{F}}(H)$. The Kronecker product $M \otimes M'$ has rows and columns indexed by $V(G) \times V(H)$, and its entry in row (g,h) and column (g',h') is $M_{gg'}M'_{hh'}$. In a field a product is nonzero exactly when both factors are, so this entry is nonzero exactly when $g' \in N[g]$ and $h' \in N[h]$, that is, when $(g',h') \in N[(g,h)]$ in $G \boxtimes H$. Thus $M \otimes M' \in \mathcal{S}_{\mathbb{F}}[G \boxtimes H]$, and since $\mathrm{rank}(M \otimes M') = \mathrm{rank}\, M \cdot \mathrm{rank}\, M'$, Lemma 2.1 gives the first bound. For the second, if $M \in \mathcal{S}^{\circ}_{\mathbb{F}}(G)$ and $M' \in \mathcal{S}^{\circ}_{\mathbb{F}}(H)$, then $M_{gg'}M'_{hh'} \neq 0$ exactly when $g' \in N(g)$ and $h' \in N(h)$, which is adjacency in $G \times H$; so $M \otimes M' \in \mathcal{S}^{\circ}_{\mathbb{F}}(G \times H)$. □

By induction, $\gamma_{\mathrm{gr}}(G_1 \boxtimes \cdots \boxtimes G_r) \le \prod_i \mathrm{mr}_{\mathbb{F}}(G_i)$ and the same for direct products, and in fact

$$\mathrm{mr}_{\mathbb{F}}(G \boxtimes H) \le \mathrm{mr}_{\mathbb{F}}(G)\ \mathrm{mr}_{\mathbb{F}}(H), \qquad \mathrm{mr}^{\circ}_{\mathbb{F}}(G \times H) \le \mathrm{mr}^{\circ}_{\mathbb{F}}(G)\ \mathrm{mr}^{\circ}_{\mathbb{F}}(H). \tag{2.3}$$

Definition 2.4. A graph G is $\mathbb{F}$-*tight* if $\gamma_{\mathrm{gr}}(G) = \mathrm{mr}_{\mathbb{F}}(G)$, and *totally* $\mathbb{F}$-*tight* if it has no isolated vertices and $\gamma^t_{\mathrm{gr}}(G) = \mathrm{mr}^{\circ}_{\mathbb{F}}(G)$. It is *tight* if it is $\mathbb{F}$-tight for some field $\mathbb{F}$.

By (2.1), an $\mathbb{F}$-tight graph is $\mathbb{F}'$-tight for every extension $\mathbb{F}'$ of $\mathbb{F}$: the minimum rank can only go down under extension, and it cannot go below $\gamma_{\mathrm{gr}}(G)$. So if G is $\mathbb{F}$-tight and H is $\mathbb{F}'$-tight, and $\mathbb{F}$ and $\mathbb{F}'$ have the same characteristic, then both are tight over a common field, a field containing copies of both.[4]

Corollary 2.5. *If $G_1, \ldots, G_r$ are $\mathbb{F}$-tight, then*

$$\gamma_{\mathrm{gr}}(G_1 \boxtimes \cdots \boxtimes G_r) = \prod_{i=1}^{r} \gamma_{\mathrm{gr}}(G_i),$$

and the product is $\mathbb{F}$-tight. If $G_1, \ldots, G_r$ are totally $\mathbb{F}$-tight, then $\gamma^t_{\mathrm{gr}}(G_1 \times \cdots \times G_r) = \prod_i \gamma^t_{\mathrm{gr}}(G_i)$ and the product is totally $\mathbb{F}$-tight. In particular, if G and H are tight over fields of the same characteristic, then G and H satisfy Conjecture 1.1.

Proof. The lower bound is (2.2). The upper bound is Theorem A with (2.3), which also gives $\mathrm{mr}_{\mathbb{F}}(\prod G_i) \le \prod \gamma_{\mathrm{gr}}(G_i) = \gamma_{\mathrm{gr}}(\prod G_i)$; with Lemma 2.1 this is an equality, so the product is tight. □

A counterexample to Conjecture 1.1 therefore needs, for every characteristic, a factor that is not tight in that characteristic. The constructions of Section 6 show how this happens.

Remark 2.6. Two vertices $u \neq v$ are *true twins* if $N[u] = N[v]$. If v is a true twin of u, then $\gamma_{\mathrm{gr}}(G - v) = \gamma_{\mathrm{gr}}(G)$, since a dominating sequence can contain at most one of u, v (after one of them is chosen the closed neighbourhood of the other is dominated) and v can be replaced by u. Duplicating the row and column of u in a closed pattern matrix of $G - v$ gives one of G with the same rank. Hence G is $\mathbb{F}$-tight if and only if $G - v$ is. In particular every graph obtained from an $\mathbb{F}$-tight graph by substituting complete graphs for vertices is $\mathbb{F}$-tight; compare [8, Proposition 30].

Justification of the last claim in Remark 2.6. If M is a closed pattern matrix of $G - v$ and M' is obtained by repeating row u as row v and column u as column v, then $M'_{vv} = M'_{uv} = M'_{vu} = M_{uu} \neq 0$, which is correct since u, v are adjacent, and for $w \neq u, v$ the entries in row and column v copy those of u, which is correct since $N[v] = N[u]$. Repeating a row and a column does not change the rank. Conversely, deleting a row and a column cannot increase the rank, so $\mathrm{mr}_{\mathbb{F}}(G - v) \le \mathrm{mr}_{\mathbb{F}}(G)$, and the inequalities $\gamma_{\mathrm{gr}}(G) = \gamma_{\mathrm{gr}}(G - v) \le \mathrm{mr}_{\mathbb{F}}(G - v) \le \mathrm{mr}_{\mathbb{F}}(G)$ show that tightness passes in both directions. □

[4] Two fields of the same characteristic embed in a common field, for instance in an algebraically closed field of that characteristic of sufficiently large transcendence degree. In all our examples a finite field or $\mathbb{C}$ suffices.

3. Tight graphs

This section produces the tight graphs used in the product formulas. In each case the lower bound for γ_{gr} is an explicit sequence and the upper bound is a matrix of the right pattern and rank.

3.1. **Complete graphs, edgeless graphs and induced subgraphs.** The all-ones matrix J is a closed pattern matrix of K_n of rank 1, and $\gamma_{\mathrm{gr}}(K_n) = 1$; the identity is a closed pattern matrix of the edgeless graph $\overline{K_n}$ of rank $n = \gamma_{\mathrm{gr}}(\overline{K_n})$. So both are tight over every field. We record a general principle for lower bounds.

Lemma 3.1. *If F is an induced subgraph of G, then every dominating sequence of F is a dominating sequence of G with the same footprints. Hence $\gamma_{\mathrm{gr}}(F) \le \gamma_{\mathrm{gr}}(G)$.*

Proof. For $v \in V(F)$ we have $N_F[v] = N_G[v] \cap V(F)$, so a vertex of F that lies in $N_F[v_i]$ and in no $N_F[v_j]$, $j < i$, lies in $N_G[v_i]$ and in no $N_G[v_j]$. □

Example 3.2. The path P_m with vertices $1, 2, \dots, m$, $m \ge 2$, has the dominating sequence $(1, 2, \dots, m-1)$: the vertex i footprints $i+1$, which no earlier vertex dominates. So $\gamma_{\mathrm{gr}}(P_m) \ge m-1$. Since a dominating sequence of a graph without isolated vertices cannot use every vertex,[5] $\gamma_{\mathrm{gr}}(P_m) = m-1$. By Lemma 3.1, a graph containing an induced path on m vertices has $\gamma_{\mathrm{gr}} \ge m-1$; in particular $\gamma_{\mathrm{gr}}(C_k) \ge k-2$.

3.2. **Paths.**

Proposition 3.3. *For $m \ge 2$ the path P_m is $\mathbb{R}$-tight.*

Proof. Let $Q = D + A$ be the signless Laplacian of P_m, where D is the diagonal matrix of degrees. Its diagonal entries are 1 or 2 and its off-diagonal entries are 1 exactly on the edges, so $Q \in \mathcal{S}_{\mathbb{R}}[P_m]$. Writing $Q = BB^{\mathsf{T}}$ with B the $m \times (m-1)$ vertex–edge incidence matrix, we have $\operatorname{rank} Q = \operatorname{rank} B \le m-1$. With Example 3.2, $m-1 = \gamma_{\mathrm{gr}}(P_m) \le \mathrm{mr}_{\mathbb{R}}(P_m) \le m-1$. □

3.3. **Complete bipartite graphs and cocktail party graphs.**

Proposition 3.4. *Let $a \ge b \ge 1$ and let $\mathbb{F}$ be a field in which $a \ne 0$ and which contains a primitive a-th root of unity. Then $K_{a,b}$ is $\mathbb{F}$-tight and $\gamma_{\mathrm{gr}}(K_{a,b}) = a$. In particular $K_{a,b}$ is $\mathbb{C}$-tight.*

Proof. Listing the a vertices of the larger side gives a dominating sequence, each vertex footprinting itself, so $\gamma_{\mathrm{gr}}(K_{a,b}) \ge a$. Let $\zeta \in \mathbb{F}$ be a primitive a-th root of unity and define the $a \times b$ matrix Z and the $b \times a$ matrix W by $Z_{kj} = \zeta^{kj}$ and $W_{jk} = \zeta^{-kj}$, for $0 \le k < a$ and $0 \le j < b$. Then

$$(WZ)_{jj'} = \sum_{k=0}^{a-1} \zeta^{k(j'-j)} = a\,\delta_{jj'},$$

because for $0 < |j'-j| < a$ the element $\zeta^{j'-j} \ne 1$ is a root of $1 + x + \cdots + x^{a-1}$. Put

$$M = \begin{pmatrix} aI_a & Z \\ W & I_b \end{pmatrix}.$$

Its nonzero entries are the diagonal and all entries between the two sides, which is the closed pattern of $K_{a,b}$, and its last b rows are $a^{-1}W$ times its first a rows, since $a^{-1}W(aI_a\ Z) = (W\ I_b)$. So $\operatorname{rank} M = a$. □

[5] The last vertex v_n of a sequence using all n vertices would need a footprint outside $N[v_1] \cup \cdots \cup N[v_{n-1}]$, which is the whole vertex set as soon as every vertex has a neighbour.

The *cocktail party graph* $CP(s) = K_{2,2,\dots,2}$, with $s \geq 2$ parts of size two, is the complete graph K_{2s} minus a perfect matching. For $s = 3$ it is the octahedron. Each vertex is adjacent to every other vertex except its *antipode*.

Proposition 3.5. *For $s \geq 2$, $\gamma_{\mathrm{gr}}(CP(s)) = 2$, and $CP(s)$ is $\mathbb{F}$-tight over every field $\mathbb{F}$ with at least $2s-1$ elements.*

Proof. A vertex u followed by its antipode $\bar{u}$ is a dominating sequence: $\bar{u}$ footprints itself. Conversely, after two vertices u, v of a sequence everything is dominated: if $v = \bar{u}$ then $N[u] \cup N[v]$ is the whole vertex set, and otherwise $N[u]$ misses only $\bar{u}$, which lies in $N[v]$. So $\gamma_{\mathrm{gr}}(CP(s)) = 2$.

For the matrix, identify $\mathbb{F}^2$ with row vectors and let $\langle x, y\rangle = x_1y_1 + x_2y_2$. For $x \in \mathbb{F}^2$ write $x^\perp = (-x_2, x_1)$, so that $\langle y, x^\perp\rangle = 0$ exactly when y is a multiple of x. Since $\mathbb{F}$ has at least $2s-1$ elements, $\mathbb{F}^2$ has $|\mathbb{F}|+1 \geq 2s$ distinct one-dimensional subspaces; choose nonzero vectors r_u, one for each of the $2s$ vertices u, spanning pairwise distinct lines. Define

$$M_{uv} = \langle r_u,\, r_{\bar{v}}^\perp\rangle.$$

Then M has rank at most 2, since $M = RC$ with R the $2s \times 2$ matrix of the rows r_u and C the $2 \times 2s$ matrix with columns $r_{\bar{v}}^{\perp\mathsf{T}}$. The entry M_{uv} vanishes exactly when r_u is a multiple of $r_{\bar{v}}$, that is, when $u = \bar{v}$; so the zero entries of M are exactly the antipodal pairs, and the diagonal entries are nonzero because $u \neq \bar{u}$. Thus $M \in \mathcal{S}_{\mathbb{F}}[CP(s)]$ and $\mathrm{mr}_{\mathbb{F}}(CP(s)) \leq 2 = \gamma_{\mathrm{gr}}(CP(s))$. □

For the octahedron $CP(3)$ this needs $|\mathbb{F}| \geq 5$; it is tight over $\mathbb{R}$, over $\mathbb{C}$ and over every finite field with at least five elements.

3.4. **Cycles.**

Proposition 3.6. *Let $k \geq 3$, $k \neq 4$, and let $\mathbb{F}$ be a field whose characteristic does not divide k and which contains a primitive k-th root of unity ζ. Then C_k is $\mathbb{F}$-tight and $\gamma_{\mathrm{gr}}(C_k) = k-2$. The same holds over $\mathbb{F} = \mathbb{R}$, and $C_4 = K_{2,2}$ is tight over every field of characteristic different from 2 that contains $-1 \neq 1$, in particular over $\mathbb{C}$. Hence every cycle is $\mathbb{C}$-tight.*

Proof. Label the vertices by $\mathbb{Z}/k$ with i adjacent to $i \pm 1$, let A be the adjacency matrix and put

$$d = -(\zeta + \zeta^{-1}), \qquad M = A + dI.$$

First, $d \neq 0$: otherwise $\zeta^2 = -1$ and $\zeta^4 = 1$, so the order k of ζ divides 4, which is excluded for $k \geq 3$, $k \neq 4$. So $M \in \mathcal{S}_{\mathbb{F}}[C_k]$. The matrix M is circulant. Let V be the $k \times k$ matrix $V_{jl} = \zeta^{jl}$; it is a Vandermonde matrix on the distinct elements $1, \zeta, \dots, \zeta^{k-1}$, hence invertible. For the column $v_j = (\zeta^{jl})_l$ of V we have

$$(Mv_j)_l = \zeta^{j(l+1)} + \zeta^{j(l-1)} + d\,\zeta^{jl} = \lambda_j\,\zeta^{jl}, \qquad \lambda_j = \zeta^j + \zeta^{-j} - \zeta - \zeta^{-1},$$

so M is diagonalised by V with eigenvalues $\lambda_0, \dots, \lambda_{k-1}$. For $x, y \neq 0$,

$$x + x^{-1} - y - y^{-1} = (x-y)(1-(xy)^{-1}),$$

so $\lambda_j = 0$ exactly when $\zeta^j = \zeta$ or $\zeta^j = \zeta^{-1}$, that is, when $j \equiv \pm 1 \pmod{k}$. These are two values of j since $k \geq 3$, and so $\operatorname{rank} M = k-2$. With Example 3.2, $k-2 \leq \gamma_{\mathrm{gr}}(C_k) \leq \mathrm{mr}_{\mathbb{F}}(C_k) \leq k-2$.

Over $\mathbb{R}$ the same computation, with $d = -2\cos(2\pi/k)$ and the real circulant $M = A + dI$, gives the eigenvalues $2\cos(2\pi j/k) - 2\cos(2\pi/k)$, which vanish exactly for $j \equiv \pm 1$; and $d \neq 0$ for $k \neq 4$. For $k = 4$ use Proposition 3.4 with $a = b = 2$; -1 is a primitive square root of unity when the characteristic is not 2. □

Remark 3.7. The fields of Proposition 3.6 include every finite field $\mathbb{F}_{p^s}$ with $p \nmid k$ and $k \mid p^s - 1$, since the multiplicative group of $\mathbb{F}_{p^s}$ is cyclic of order $p^s - 1$. So for each odd k the cycle C_k is tight over a finite field of characteristic 2, which is the case used for Kneser graphs below.

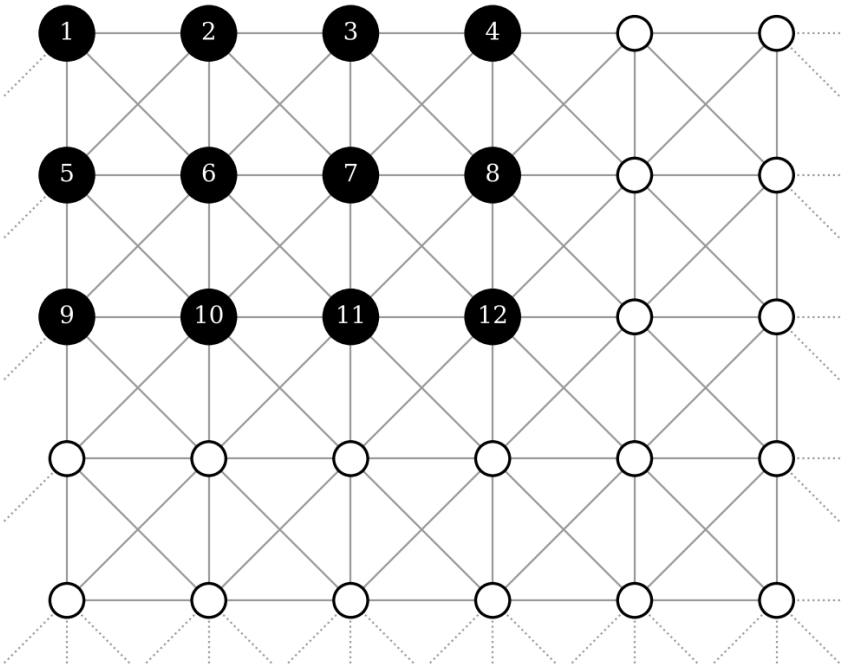


FIGURE 3. The strong torus $C_5 \boxtimes C_6$, drawn as a 5×6 grid whose opposite sides are identified (dotted edges wrap around). The $12 = (5-2)(6-2)$ marked vertices form a longest dominating sequence; by Theorem B no dominating sequence is longer.

In characteristic $p \mid k$, the polynomial $x^k - 1$ has repeated roots, the circulant is no longer diagonalisable, and the argument does not apply.

3.5. **Kneser graphs.** The Kneser graph $K(n,2)$ has the two-element subsets of $[n] = \{1, \dots, n\}$ as vertices, two of them adjacent when they are disjoint. For $n = 5$ it is the Petersen graph.

Proposition 3.8. (i) *For $n \geq 6$, $\gamma_{\mathrm{gr}}(K(n,2)) = n - 1$, and $\gamma_{\mathrm{gr}}(K(5,2)) = 5$.*

(ii) *If $n \geq 6$ is even, $K(n,2)$ is $\mathbb{F}_2$-tight.*

(iii) *If $n \geq 7$ is odd, $K(n,2)$ is $\mathbb{F}_p$-tight for every odd prime p dividing $n - 4$; such a prime exists.*

(iv) *The Petersen graph $K(5,2)$ is $\mathbb{F}$-tight over every field of characteristic different from 2.*

Proof. Lower bounds. The $n - 1$ sets $\{1, j\}$, $2 \leq j \leq n$, pairwise intersect, so they form an independent set, and listing them gives a dominating sequence in which each set footprints itself. For $n = 5$ the sequence

$$\{1,2\},\ \{1,3\},\ \{2,4\},\ \{1,5\},\ \{1,4\}$$

has the footprints $\{1,2\}$, $\{1,3\}$, $\{1,5\}$, $\{2,3\}$, $\{1,4\}$, as one checks from the rule that a set dominates itself and the three sets disjoint from it.

The matrix identity. Let U be the $\binom{n}{2} \times n$ incidence matrix, $U_{S,x} = 1$ if $x \in S$. Over the integers,

$$(UU^{\mathsf{T}})_{ST} = |S \cap T| = \begin{cases} 2, & S = T, \\ 1, & |S \cap T| = 1, \\ 0, & S \cap T = \emptyset. \end{cases}$$

Writing A_0 for the adjacency matrix of $K(n,2)$ and A_1 for the matrix of the relation $|S \cap T| = 1$, we have $UU^{\mathsf{T}} = 2I + A_1$ and $J = I + A_0 + A_1$, hence

$$A_0 = J + I - UU^{\mathsf{T}}. \tag{3.1}$$

(iii) and (iv). Let p be an odd prime, or let $\mathbb{F}$ have characteristic different from 2. Every row of U has two ones, so $U\mathbf{1}_n = 2 \cdot \mathbf{1}$, and $J = \mathbf{1}\mathbf{1}^{\mathsf{T}} = \frac{1}{4}U\mathbf{1}_n\mathbf{1}_n^{\mathsf{T}}U^{\mathsf{T}} = \frac{1}{4}UJ_nU^{\mathsf{T}}$. With (3.1),

$$A_0 - I = J - UU^{\mathsf{T}} = U(\tfrac{1}{4}J_n - I_n)U^{\mathsf{T}}.$$

The matrix $A_0 - I$ has diagonal entries $-1 \neq 0$, entries 1 on disjoint pairs and 0 on intersecting pairs, so $A_0 - I \in \mathcal{S}_{\mathbb{F}}[K(n,2)]$, and $\operatorname{rank}(A_0 - I) \leq \operatorname{rank}(\frac{1}{4}J_n - I_n) \leq n$. For $n = 5$ this gives $\operatorname{mr}_{\mathbb{F}}(K(5,2)) \leq 5 = \gamma_{\mathrm{gr}}(K(5,2))$, which is (iv). For (iii), let $n \geq 7$ be odd and p an odd prime dividing $n-4$; one exists because $n-4 \geq 3$ is odd. Then $n \equiv 4 \not\equiv 0 \pmod{p}$, so $\mathbb{F}_p^n$ is the direct sum of the line spanned by $\mathbf{1}_n$ and the hyperplane of vectors with coordinate sum 0. The matrix $\frac{1}{4}J_n - I_n$ acts on the line by $\frac{n}{4} - 1 = 0$ and on the hyperplane by -1. So its rank is $n-1$, and $\operatorname{mr}_{\mathbb{F}_p}(K(n,2)) \leq n - 1 = \gamma_{\mathrm{gr}}(K(n,2))$.

(ii). Let $n \geq 6$ be even and work over $\mathbb{F}_2$. Reducing (3.1) modulo 2 gives $A_0 + I = J + UU^{\mathsf{T}} = WW^{\mathsf{T}}$, where $W = (U \mid \mathbf{1})$ is the $\binom{n}{2} \times (n+1)$ matrix obtained by appending a column of ones. The matrix $A_0 + I$ lies in $\mathcal{S}_{\mathbb{F}_2}[K(n,2)]$. For any matrix W,

$$\operatorname{rank}(WW^{\mathsf{T}}) = \dim W(\operatorname{im} W^{\mathsf{T}}) = \operatorname{rank} W - \dim\bigl(\operatorname{im} W^{\mathsf{T}} \cap \ker W\bigr).$$

The vector $z = (\mathbf{1}_n, 0) \in \mathbb{F}_2^{n+1}$ lies in $\ker W$, since $Wz = U\mathbf{1}_n = 2 \cdot \mathbf{1} = 0$; this also shows $\operatorname{rank} W \leq n$. We claim that $z \in \operatorname{im} W^{\mathsf{T}}$, the row space of W. The row of W indexed by $S = \{x, y\}$ is $(e_x + e_y, 1)$, so the sum of the rows over the edge set E of a graph Γ on $[n]$ is $(\sum_x \deg_\Gamma(x) e_x,\ |E|)$ modulo 2. It suffices to find a graph Γ on $[n]$ in which every vertex has odd degree and the number of edges is even. If $n \equiv 0 \pmod 4$, a perfect matching has $n/2$ edges, an even number. If $n \equiv 2 \pmod 4$, take a star with centre 1 and leaves $2, 3, 4$ together with a perfect matching of $\{5, \dots, n\}$; this has $3 + (n-4)/2$ edges, which is even because $(n-4)/2$ is odd. Hence $\dim(\operatorname{im} W^{\mathsf{T}} \cap \ker W) \geq 1$ and $\operatorname{rank}(A_0 + I) \leq n - 1 = \gamma_{\mathrm{gr}}(K(n,2))$.

Upper bound in (i). For $n \geq 6$, (ii) and (iii) give $\gamma_{\mathrm{gr}}(K(n,2)) \leq \operatorname{mr}_{\mathbb{F}}(K(n,2)) \leq n - 1$ for a suitable field. For $n = 5$, (iv) gives $\gamma_{\mathrm{gr}} \leq 5$. □

The equality $\gamma_{\mathrm{gr}}(K(n,2)) = n - 1$ for $n \geq 6$ was proved by Brešar, Kos and Torres [13] by a combinatorial argument; the matrices above give a two-line upper bound once the identity (3.1) is written down. The value 5 for the Petersen graph agrees with Lin's computation [18, Example 3.2] over the reals.

Remark 3.9. For even n the field $\mathbb{F}_2$ cannot be replaced by a field of characteristic 0: over $\mathbb{Q}$ the matrix $A_0 - I$ has rank n, since $\frac{1}{4}J_n - I_n$ is invertible when $n \neq 4$, and every matrix $aA_0 + bI$ in the Bose–Mesner algebra of the Johnson scheme has rank at least n for $n \geq 6$. We do not know whether $\operatorname{mr}_{\mathbb{C}}(K(n,2)) = n - 1$ for some $n \geq 6$.

4. Kneser graphs of every rank

For $n \geq 2r \geq 2$ the Kneser graph $K(n,r)$ has the r-element subsets of $[n]$ as vertices, two of them adjacent when they are disjoint. By the Erdős–Ko–Rado theorem [15], $\alpha(K(n,r)) = \binom{n-1}{r-1}$ for $n \geq 2r+1$, attained by the star of all r-sets containing a fixed point. An independent set listed in any order is a dominating sequence, each vertex footprinting itself, so together with Lemma 2.1 we have, for every graph G, every field $\mathbb{F}$ and every $M \in \mathcal{S}_{\mathbb{F}}[G]$,

$$\alpha(G) \leq \gamma_{\mathrm{gr}}(G) \leq \mathrm{mr}_{\mathbb{F}}(G) \leq \operatorname{rank} M. \tag{4.1}$$

When the two ends agree, all four numbers are equal. Brešar, Kos and Torres [13] proved $\gamma_{\mathrm{gr}}(K(n,2)) = \alpha(K(n,2))$ for $n \geq 6$ and, for each $r \geq 3$, $\gamma_{\mathrm{gr}}(K(n,r)) = \alpha(K(n,r))$ for all n larger than an unspecified threshold $\bar{n}_r$; they also found $\gamma_{\mathrm{gr}}(K(5,2)) = 5$ and $\gamma_{\mathrm{gr}}(K(7,3)) = 20$, so the threshold exceeds $2r+1$ in general. Campêlo and Severín [14] computed further cases and conjectured the exact threshold. In this section we prove their conjecture for all but finitely many n for each r, by a matrix over a prime field that extends the matrix used for $K(n,2)$ in Proposition 3.8.

4.1. **Matrices on r-sets.** Let $[n] = \{1, \dots, n\}$ and let $\binom{[n]}{r}$ be the set of r-subsets of $[n]$. For $0 \leq i \leq r$ let W_i be the inclusion matrix with rows indexed by $\binom{[n]}{i}$ and columns by $\binom{[n]}{r}$, $(W_i)_{U,S} = 1$ if $U \subseteq S$ and 0 otherwise. Then

$$(W_i^{\mathsf{T}} W_i)_{S,T} = \binom{|S \cap T|}{i}, \tag{4.2}$$

the number of i-subsets of $S \cap T$. Consequently, if f is a polynomial with integer values written as $f(t) = \sum_{i=0}^{d} c_i \binom{t}{i}$ with $c_i \in \mathbb{Z}$, then the integer matrix $\big(f(|S \cap T|)\big)_{S,T}$ equals $\sum_{i \leq d} c_i W_i^{\mathsf{T}} W_i$, and for every prime p

$$\operatorname{rank}_{\mathbb{F}_p} \big(f(|S \cap T|) \bmod p\big)_{S,T} \leq \sum_{i=0}^{d} \binom{n}{i}. \tag{4.3}$$

Every polynomial with integer coefficients has this form, because $t^j = \sum_i i!\, S(j,i) \binom{t}{i}$ with the Stirling numbers $S(j,i)$ of the second kind.

The generalized binomial coefficient $\binom{x}{k} = x(x-1)\cdots(x-k+1)/k!$ is defined for all integers x and $k \geq 0$; it is an integer, and $\binom{-1}{k} = (-1)^k$.

Lemma 4.1. *For integers t and $k \geq 0$,*

$$\sum_{i=0}^{k} (-1)^i \binom{t}{i} = (-1)^k \binom{t-1}{k}.$$

Proof. For $k = 0$ both sides are 1. If the identity holds for $k-1$, then by Pascal's rule $\binom{t-1}{k} + \binom{t-1}{k-1} = \binom{t}{k}$ the left side for k is $(-1)^{k-1}\binom{t-1}{k-1} + (-1)^k \binom{t}{k} = (-1)^k \binom{t-1}{k}$. □

Lemma 4.2. *Let p be a prime, $a \geq 1$, $Q \geq 0$ an integer divisible by p^a, and $0 \leq k < p^a$. Then $\binom{Q+k}{k} \equiv 1 \pmod{p}$.*

Proof. Write $\binom{Q+k}{k} = \prod_{j=1}^{k} \frac{Q+j}{j}$. For $1 \leq j \leq k$ let p^b be the exact power of p dividing j, so $b < a$, and $j = p^b j'$ with $p \nmid j'$. Then $Q + j = p^b(Q/p^b + j')$ and $Q/p^b \equiv 0 \pmod{p}$, so $\frac{Q+j}{j} = \frac{Q/p^b + j'}{j'}$ is a quotient of two integers prime to p that are congruent modulo p. The product is an integer, and in the local ring $\mathbb{Z}_{(p)}$ it is a product of units congruent to 1 modulo p; so it is congruent to 1 modulo p. □

Lemma 4.2 is the case of Lucas's theorem [19] that we need; the condition $k < p^a$ cannot be dropped, since $\binom{p^a+p^a}{p^a} \equiv 2 \pmod p$.

4.2. **The threshold.**

Conjecture 4.3 (Campêlo and Severín [14, Section 6.5])**.** *For all $r \geq 2$ and $n \geq 3r$, $\gamma_{\mathrm{gr}}(K(n,r)) = \binom{n-1}{r-1}$.*

For $n \geq 2r$ put $m = n - 2r$ and

$$L_r = \operatorname{lcm}(1, 2, \ldots, r-1) \qquad (L_1 = L_2 = 1,\ L_3 = 2,\ L_4 = 6,\ L_5 = 12,\ L_6 = 60).$$

Theorem 4.4. *Let $r \geq 2$ and $n \geq 2r+1$. If $n - 2r$ does not divide L_r, then*

$$\gamma_{\mathrm{gr}}(K(n,r)) = \alpha(K(n,r)) = \binom{n-1}{r-1}.$$

More precisely, if a prime power $p^a \geq r$ divides $n - 2r$, then the matrix

$$M_{n,r} = \Big(\binom{|S \cap T| - 1}{r-1} \bmod p\Big)_{S,T \in \binom{[n]}{r}}$$

is a closed pattern matrix of $K(n,r)$ over $\mathbb{F}_p$ and has rank $\binom{n-1}{r-1}$. So $K(n,r)$ is $\mathbb{F}_p$-tight.

The two hypotheses are equivalent: $m = n-2r \geq 1$ has no prime-power divisor $p^a \geq r$ exactly when $p^{v_p(m)} \leq r-1$ for every prime p, that is, when $p^{v_p(m)}$ divides L_r for every p, that is, when $m \mid L_r$. Before the proof we record what the theorem gives.

Corollary 4.5. *Let $r \geq 2$.*

(a) *Conjecture 4.3 holds for all $n \geq 3r$ except possibly the finitely many n with $n - 2r \mid L_r$ and $n - 2r \geq r$. In particular it holds for every $n > 2r + L_r$, so the threshold of Brešar, Kos and Torres satisfies $\bar{n}_r \leq 2r + L_r + 1$.*

(b) *Conjecture 4.3 holds for $r = 2$ and $r = 3$ without exception, for $r = 4$ except possibly $n = 14$, for $r = 5$ except possibly $n \in \{16, 22\}$, and for $r = 6$ except possibly $n \in \{18, 22, 24, 27, 32, 42, 72\}$.*

(c) *If r is a prime power, Conjecture 4.3 holds at the threshold $n = 3r$.*

(d) *For $n \geq 2r+1$ with $n - 2r \nmid L_r$ the Kneser graph has $\gamma_{\mathrm{gr}} = \alpha$, so it lies in the class $\mathcal{F}_\alpha$ studied in* [2]*, and its minimum rank over the prime field of the theorem is $\mathrm{mr}_{\mathbb{F}_p}(K(n,r)) = \binom{n-1}{r-1}$.*

Proof. (a) If $n \geq 3r$ and $m = n - 2r \nmid L_r$, Theorem 4.4 applies. The divisors of L_r are finite in number and at most L_r. (b) $L_2 = 1$: the only exception is $m = 1$, $n = 5 < 6$. $L_3 = 2$: the exceptions $m \in \{1, 2\}$ give $n \in \{7, 8\}$, below $3r = 9$. $L_4 = 6$: $m \in \{1, 2, 3, 6\}$ gives $n \in \{9, 10, 11, 14\}$, and only $14 \geq 12$. $L_5 = 12$: $m \in \{1, 2, 3, 4, 6, 12\}$ gives $n \in \{11, 12, 13, 14, 16, 22\}$. $L_6 = 60$: the divisors $m \geq 6$ of 60 are $6, 10, 12, 15, 20, 30, 60$, giving $n \in \{18, 22, 24, 27, 32, 42, 72\}$. (c) For $n = 3r$ we have $m = r$, and if $r = p^a$ then $p^a \geq r$ divides m. (d) The sandwich (4.1) with $M = M_{n,r}$. □

Table 1 lists the undecided values for $r \leq 8$. The density of the exceptional set is zero for each r, and, since a prime power $p^a \geq r$ dividing $n - 2r$ forces $n - 2r \geq r$, the theorem never applies below $3r$; this is consistent with the values $\gamma_{\mathrm{gr}}(K(5,2)) = 5$, $\gamma_{\mathrm{gr}}(K(7,3)) = 20$ and Theorem 4.9 below, all of which exceed α. The smallest admissible n is $2r + q$, with q the least prime power $\geq r$.

r	L_r	$3r$	$n \geq 3r$ not covered by Theorem 4.4
2	1	6	none
3	2	9	none
4	6	12	14
5	12	15	16, 22
6	60	18	18, 22, 24, 27, 32, 42, 72
7	60	21	24, 26, 29, 34, 44, 74
8	420	24	26, 28, 30, 31, 36, 37, 44, 46, 51, 58, 76, 86, 100, 121, 156, 226, 436

TABLE 1. The values $n \geq 3r$ with $n - 2r \mid L_r$, for which Conjecture 4.3 is not decided by Theorem 4.4. The case $(n, r) = (14, 4)$ is discussed in Proposition 4.10.

4.3. **The congruence.** Fix $r \geq 2$ and put

$$g(t) = \binom{t-1}{r-1} \qquad (t \in \mathbb{Z}).$$

For $0 \leq t \leq r$ this gives

$$g(r) = 1, \qquad g(t) = 0 \ \ (1 \leq t \leq r-1), \qquad g(0) = \binom{-1}{r-1} = (-1)^{r-1}, \tag{4.4}$$

so for every prime p the matrix $M_{n,r}$ has a nonzero entry in position (S, T) exactly when $|S \cap T| \in \{0, r\}$, that is, when $T \in N[S]$ in $K(n, r)$. Put $\varepsilon = (-1)^{r-1}$. By Lemma 4.1 with $k = r - 1$,

$$g(t) = \varepsilon \sum_{i=0}^{r-1} (-1)^i \binom{t}{i}. \tag{4.5}$$

Hence, for any finite sets A and R,

$$g(|A \cap R|) = \varepsilon \sum_{\substack{U \subseteq R \\ |U| \leq r-1}} (-1)^{|U|} \, [U \subseteq A], \tag{4.6}$$

where $[\cdot]$ is 1 if the statement holds and 0 otherwise; indeed the number of i-subsets U of R with $U \subseteq A$ is $\binom{|A \cap R|}{i}$.

Proposition 4.6. *Let p be a prime and $a \geq 1$ with $p^a \geq r$ and $p^a \mid n - 2r$. Fix $x \in [n]$. Then for every $T \in \binom{[n]}{r}$ with $x \notin T$ and every $R \in \binom{[n]}{r}$,*

$$g(|T \cap R|) \equiv \varepsilon \sum_{\substack{S \in \binom{[n]}{r} \\ x \in S, \ S \cap T = \emptyset}} g(|S \cap R|) \pmod{p}. \tag{4.7}$$

Proof. Insert (4.6) with $A = S$ into the right side and exchange the sums. Since $\varepsilon^2 = 1$, the right side becomes

$$\sum_{\substack{U \subseteq R \\ |U| \leq r-1}} (-1)^{|U|} \, c(U), \qquad c(U) = \#\Big\{ S \in \tbinom{[n]}{r} : x \in S, \ S \cap T = \emptyset, \ U \subseteq S \Big\}.$$

If U meets T then $c(U) = 0$. Otherwise put $U^* = U \cup \{x\}$ and $u = |U^*|$; then $1 \leq u \leq r$, because $|U| \leq r - 1$. The sets counted by $c(U)$ are the r-sets with $U^* \subseteq S \subseteq [n] \setminus T$, and $|[n] \setminus T| = n - r$,

so

$$c(U)=\binom{n-r-u}{r-u}=\binom{(n-2r)+(r-u)}{r-u}\equiv 1 \pmod p$$

by Lemma 4.2, applied with $Q=n-2r$ and $k=r-u\le r-1<p^a$. Therefore the right side of (4.7) is congruent to

$$\Sigma=\sum_{\substack{U\subseteq R\setminus T\\ |U|\le r-1}}(-1)^{|U|}=\sum_{j=0}^{\min(k,r-1)}(-1)^j\binom{k}{j},\qquad k=|R\setminus T|=r-|R\cap T|.$$

If $k\le r-1$, that is, if R meets T, then $\Sigma=(1-1)^k=[k=0]=[R=T]$. If $k=r$, that is, if $R\cap T=\emptyset$, then $\Sigma=(1-1)^r-(-1)^r=(-1)^{r-1}=\varepsilon$. By (4.4) the left side of (4.7) takes exactly these values: 1 if $R=T$, 0 if $1\le|R\cap T|\le r-1$, and ε if $R\cap T=\emptyset$. □

Proof of Theorem 4.4. Let $p^a\ge r$ divide $n-2r$ and fix $x\in[n]$. By (4.4) the matrix $M_{n,r}$ lies in $\mathcal{S}_{\mathbb{F}_p}[K(n,r)]$. Proposition 4.6 says that the row of $M_{n,r}$ indexed by any $T\not\ni x$ is ε times the sum of the rows indexed by the sets $S\ni x$ disjoint from T. So the $\binom{n-1}{r-1}$ rows indexed by the star of x span the row space, and $\operatorname{rank}_{\mathbb{F}_p}M_{n,r}\le\binom{n-1}{r-1}$. By the sandwich (4.1),

$$\binom{n-1}{r-1}=\alpha(K(n,r))\le\gamma_{\mathrm{gr}}(K(n,r))\le\operatorname{rank}_{\mathbb{F}_p}M_{n,r}\le\binom{n-1}{r-1},$$

and all four quantities are equal. □

Remark 4.7. For $r=2$ the hypothesis says that some prime p divides $n-4$, and $M_{n,2}=\big((|S\cap T|-1)\bmod p\big)_{S,T}$ is $I-A_0$ in the notation of Proposition 3.8. For $p=2$ this is the matrix A_0+I of part (ii) and for odd p the matrix $-(A_0-I)$ of part (iii) and of Proposition 5.2; so Theorem 4.4 contains these statements, with a different proof of the rank bound.

Remark 4.8. Modulo p the matrix $M_{n,r}$ is $I+\varepsilon D$, where D is the adjacency matrix of $K(n,r)$, by (4.4). Over $\mathbb{Q}$ the matrix $I+\varepsilon D$ is far from having rank $\binom{n-1}{r-1}$. The eigenvalues of D are $\lambda_j=(-1)^j\binom{n-r-j}{r-j}$ for $0\le j\le r$, with multiplicities $\binom{n}{j}-\binom{n}{j-1}$ [17], so $I+\varepsilon D$ has the eigenvalues $1+(-1)^{r-1+j}\binom{n-r-j}{r-j}$. For $j=r$ this is 0; for $j<r$ it is nonzero as soon as $n\ge 2r+2$, and then the rational rank is $\binom{n}{r}-\big(\binom{n}{r}-\binom{n}{r-1}\big)=\binom{n}{r-1}$, larger than $\binom{n-1}{r-1}$ by $\binom{n-1}{r-2}$. The hypothesis $p^a\mid n-2r$ makes every $\binom{n-r-j}{r-j}$ congruent to 1, and so every eigenvalue with $j\equiv r \pmod 2$ congruent to 0, modulo p; the number of r-sets in the star is

$$\binom{n-1}{r-1}=\sum_{\substack{0\le j\le r-1\\ j\equiv r-1 \pmod 2}}\Big(\binom{n}{j}-\binom{n}{j-1}\Big),$$

the total multiplicity of the surviving eigenvalues. The eigenvalue heuristic is not a proof in characteristic p, where the eigenspaces need not reduce well; Proposition 4.6 replaces it by an explicit linear relation.

4.4. **Small Kneser graphs.** For the values of n not covered by Theorem 4.4 the same matrix still gives upper bounds. Write $D_{n,r}$ for the adjacency matrix of $K(n,r)$ and put

$$\beta(n,r)=\min\big\{\operatorname{rank}_{\mathbb{F}_p}(I+cD_{n,r}):p\in\{2,3,5,7,11,13\},\ c\in\mathbb{F}_p^\times\big\}.$$

Every matrix $I+cD_{n,r}$ lies in $\mathcal{S}_{\mathbb{F}_p}[K(n,r)]$, so $\gamma_{\mathrm{gr}}(K(n,r))\le\beta(n,r)$.

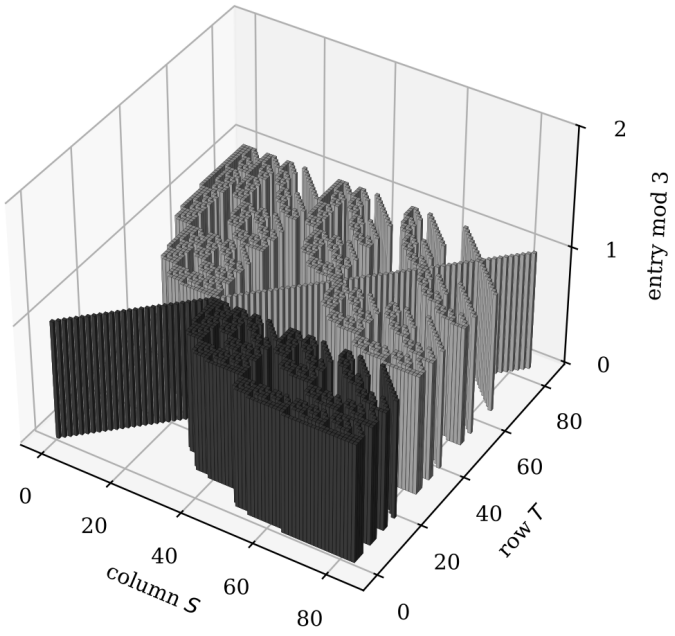


FIGURE 4. The matrix $M_{9,3}$ of Theorem 4.4 over $\mathbb{F}_3$, with the rows and columns of the star of a point first. Heights are the entries modulo 3; the dark rows, indexed by the star, span the row space (Proposition 4.6).

Theorem 4.9. $\gamma_{\mathrm{gr}}(K(8,3)) = 22$, $\gamma_{\mathrm{gr}}(K(9,4)) = 77$, $\gamma_{\mathrm{gr}}(K(10,4)) = 93$, $\gamma_{\mathrm{gr}}(K(11,4)) = 121$ *and* $\gamma_{\mathrm{gr}}(K(11,5)) = 296$.

Proof. The upper bounds are the ranks

$$\mathrm{rank}_{\mathbb{F}_2}(I + D_{8,3}) = 22, \quad \mathrm{rank}_{\mathbb{F}_3}(I + 2D_{9,4}) = 77, \quad \mathrm{rank}_{\mathbb{F}_2}(I + D_{10,4}) = 93,$$

$$\mathrm{rank}_{\mathbb{F}_3}(I + 2D_{11,4}) = 121, \quad \mathrm{rank}_{\mathbb{F}_3}(I + D_{11,5}) = 296.$$

The lower bounds are dominating sequences of these lengths, listed in Appendix B together with a footprint for every term. That the five sequences are dominating sequences is checked by the Lean kernel, and the ranks are computed by Gaussian elimination over $\mathbb{F}_p$ in two independent implementations (Appendix C). □

The previously known bounds were $21 \le \gamma_{\mathrm{gr}}(K(8,3)) \le 37$, $56 \le \gamma_{\mathrm{gr}}(K(9,4)) \le 115$, $84 \le \gamma_{\mathrm{gr}}(K(10,4)) \le 179$, $120 \le \gamma_{\mathrm{gr}}(K(11,4)) \le 265$ and $210 \le \gamma_{\mathrm{gr}}(K(11,5)) \le 448$, from the independence number and the upper bounds of [14, Table 5], where dominating sequences of lengths 22, 77, 93, 121 and 296 were found by tabu search. Theorem 4.9 shows that all five of these sequences are optimal.

Proposition 4.10. $286 \le \gamma_{\mathrm{gr}}(K(14,4)) \le 287$.

Proof. The lower bound is α; the upper bound is $\mathrm{rank}_{\mathbb{F}_3}(I + 2D_{14,4}) = 287$. □

So $(n,r) = (14,4)$, the first case of Conjecture 4.3 not covered by Theorem 4.4, is decided by a single sequence: the conjecture holds there if and only if there is no dominating sequence of length 287. In every case computed so far, $\beta(n,r) = \gamma_{\mathrm{gr}}(K(n,r))$ whenever both are known (Table 2).

Table 2 lists every Kneser graph with $n \geq 2r+1$, $3 \leq r \leq 5$ and at most 5000 vertices that is not covered by Theorem 4.4; the value $\gamma_{\mathrm{gr}}(K(7,3)) = 20$ is due to [13]. The column "known" gives the bounds of [14, Table 5]; the column β is the least rank found among the matrices $I + cD_{n,r}$ over the listed field, and "seq" is the longest dominating sequence we found. For $K(13,5)$, $K(14,5)$ and $K(16,5)$ the randomized search fell short of α (it reached 479, 648 and 1365), so the lower bound recorded is α.

TABLE 2. Kneser graphs not covered by Theorem 4.4. $\mathbb{F}_p$, c: field and scalar giving $\beta = \operatorname{rank}(I + cD_{n,r})$. Bold: exact value.

n	r	$\lvert V\rvert$	α	known	seq	β	$\mathbb{F}_p$, c
7	3	35	15	**20**	20	20	$\mathbb{F}_3$, 1
8	3	56	21	22–37	22	**22**	$\mathbb{F}_2$, 1
9	4	126	56	77–115	77	**77**	$\mathbb{F}_3$, 2
10	4	210	84	93–179	93	**93**	$\mathbb{F}_2$, 1
11	4	330	120	121–265	121	**121**	$\mathbb{F}_3$, 2
14	4	1001	286	—	286	287	$\mathbb{F}_3$, 2
11	5	462	210	296–448	296	**296**	$\mathbb{F}_3$, 1
12	5	792	330	379–746	377	384	$\mathbb{F}_2$, 1
13	5	1287	495	—	α	507	$\mathbb{F}_3$, 1
14	5	2002	715	—	α	716	$\mathbb{F}_2$, 1
16	5	4368	1365	—	1365	1380	$\mathbb{F}_3$, 1

So $379 \leq \gamma_{\mathrm{gr}}(K(12,5)) \leq 384$, $495 \leq \gamma_{\mathrm{gr}}(K(13,5)) \leq 507$, $715 \leq \gamma_{\mathrm{gr}}(K(14,5)) \leq 716$ and $1365 \leq \gamma_{\mathrm{gr}}(K(16,5)) \leq 1380$. The upper bound 384 for $K(12,5)$ halves the previous one, 746. The case $K(16,5)$ is one of the two open cases of Conjecture 4.3 for $r = 5$.

4.5. **Z-Grundy domination and zero forcing.** In the Z-variant the footprint of v_i lies in the open neighbourhood $N(v_i)$ and outside the closed neighbourhoods of the earlier terms. The Z-Grundy number is $|V(G)| - Z(G)$, where $Z(G)$ is the zero forcing number [9]. Brešar, Kos and Torres [13] proved $\gamma^Z_{\mathrm{gr}}(K(n,r)) = \binom{2r}{r}$ for $n \geq 3r+1$ and $\binom{2r}{r} - \binom{4r-1-n}{3r-n} \leq \gamma^Z_{\mathrm{gr}}(K(n,r)) \leq \binom{2r}{r}$ for $2r+1 \leq n \leq 3r$, and remarked that it is unclear whether $\gamma^Z_{\mathrm{gr}}(K(n,r))$ depends on n in the second range [13, Section 7]. The rank method applies with a free diagonal.

Lemma 4.11. *Let M be a matrix over a field with $M_{uv} \neq 0$ for adjacent u, v and $M_{uv} = 0$ for distinct nonadjacent u, v, the diagonal being arbitrary. Then $\gamma^Z_{\mathrm{gr}}(G) \leq \operatorname{rank} M$.*

Proof. Let $(v_1, \ldots, v_k)$ be a Z-sequence with footprints $w_i \in N(v_i)$, $w_i \notin N[v_j]$ for $j < i$. Then $M_{v_i w_i} \neq 0$ because w_i is a neighbour of v_i, and $M_{v_j w_i} = 0$ for $j < i$ because w_i is neither v_j nor a neighbour of v_j. The submatrix on rows $v_1, \ldots, v_k$ and columns $w_1, \ldots, w_k$ is triangular with nonzero diagonal. □

Proposition 4.12. *$\gamma^Z_{\mathrm{gr}}(K(5,2)) = \gamma^Z_{\mathrm{gr}}(K(6,2)) = 5$, while $\gamma^Z_{\mathrm{gr}}(K(n,2)) = 6$ for $n \geq 7$. Hence $Z(K(6,2)) = 10$, the lower bound of* [13] *is attained at $n = 6$ and not at $n = 5$, and for $r = 2$ the Z-Grundy number is constant on the range $2r+1 \leq n \leq 3r$ and smaller there than its value $\binom{2r}{r}$ for $n \geq 3r+1$.*

Proof. The matrix $A_0 + I$ of $K(5,2)$ and of $K(6,2)$ has rank 5 over $\mathbb{F}_2$, and its off-diagonal pattern is the adjacency of the graph, so $\gamma^Z_{\mathrm{gr}} \leq 5$ by Lemma 4.11. The lower bound $\binom{4}{2} - \binom{7-n}{6-n}$ of [13] is 5 for $n = 6$. For the Petersen graph $K(5,2)$ the vertices $\{1,2\}, \{1,3\}, \{2,4\}, \{1,5\}, \{2,3\}$,

in this order, form a Z-sequence with footprints $\{3,4\},\{2,4\},\{1,5\},\{2,3\},\{1,4\}$: each footprint is disjoint from its own term and meets every earlier term without being equal to it. So $\gamma_{\mathrm{gr}}^{Z}(K(5,2)) \geq 5$. The value 6 for $n \geq 7$ is the theorem of [13], and $Z(K(6,2)) = 15-5$. □

For $r=3$ and $r=4$ the least rank of $D_{n,r}+cI$ over the primes up to 11 is $\binom{2r}{r}$ for $n=2r+1$ and larger than $\binom{2r}{r}$ for $2r+2 \leq n \leq 3r$, so the rank method does not decide these cases.

5. Exact values for strong products

Proof of Theorem B. By Propositions 3.3, 3.4 and 3.6, every path, cycle and complete bipartite graph is $\mathbb{C}$-tight, and so is every complete graph. Corollary 2.5 gives the product formula for any strong product of such graphs, and in particular

$$\gamma_{\mathrm{gr}}\Big(\boxtimes_{i=1}^{r} C_{k_i}\Big) = \prod_{i=1}^{r} \gamma_{\mathrm{gr}}(C_{k_i}) = \prod_{i=1}^{r} (k_i - 2).$$

For Kneser graphs, let $n \geq 6$ be even and $k \geq 3$ odd. Choose s with $k \mid 2^s - 1$; then $\mathbb{F}_{2^s}$ contains $\mathbb{F}_2$ and a primitive k-th root of unity, so $K(n,2)$ (Proposition 3.8(ii) and (2.1)) and C_k (Proposition 3.6) are both $\mathbb{F}_{2^s}$-tight, and Corollary 2.5 gives $\gamma_{\mathrm{gr}}(K(n,2) \boxtimes C_k) = (n-1)(k-2)$. If $n, m \geq 6$ are even, both Kneser graphs are $\mathbb{F}_2$-tight and $\gamma_{\mathrm{gr}}(K(n,2) \boxtimes K(m,2)) = (n-1)(m-1)$. □

The proof gives more. We collect the cases in one statement.

Corollary 5.1. *Let $G_1, \dots, G_r$ be graphs each of which is a complete graph, an edgeless graph, a path, a cycle, a complete bipartite graph or a cocktail party graph. Then $\gamma_{\mathrm{gr}}(G_1 \boxtimes \cdots \boxtimes G_r) = \prod_i \gamma_{\mathrm{gr}}(G_i)$, where*

$$\gamma_{\mathrm{gr}}(K_n) = 1, \qquad \gamma_{\mathrm{gr}}(\overline{K_n}) = n, \qquad \gamma_{\mathrm{gr}}(P_m) = m-1,$$
$$\gamma_{\mathrm{gr}}(C_k) = k-2, \qquad \gamma_{\mathrm{gr}}(K_{a,b}) = \max(a,b), \qquad \gamma_{\mathrm{gr}}(CP(s)) = 2.$$

The same holds if some of the factors are Petersen graphs. It also holds for factors $K(n,2)$ with n even together with cycles of odd length and complete graphs, and for factors $K(n,2)$ with n odd together with cycles C_k such that some odd prime p divides $n-4$ for every such n and does not divide any such k.

Proof. All the graphs in the first list are $\mathbb{C}$-tight, by the propositions of Section 3; the Petersen graph is $\mathbb{C}$-tight by Proposition 3.8(iv). For the Kneser factors with n even use a finite field of characteristic 2 containing primitive roots of unity of all the odd cycle lengths, and for n odd a finite field of characteristic p containing primitive roots of unity of all the cycle lengths, which exists because p divides none of them. □

Table 3 lists some of the values. The torus entries settle the case left open in [8, Corollary 29], where the bound $(k-2)(\ell-1)$ was proved; the cylinder and grid entries recover [8, Corollaries 24 and 28].

For $K(7,2)$ and $K(10,2)$ the common odd prime is 3, which divides $7-4$ and $10-4$. That the even graph $K(10,2)$ is also tight over $\mathbb{F}_3$ follows from the next proposition, whose proof is the argument for part (iii) of Proposition 3.8.

Proposition 5.2. *Let $n \geq 6$ and let p be an odd prime with $n \equiv 4 \pmod p$. Then $K(n,2)$ is $\mathbb{F}_p$-tight.*

Proof. The proof of Proposition 3.8(iii) used only that p is odd and $n \equiv 4 \not\equiv 0 \pmod p$. □

TABLE 3. Grundy domination numbers of strong products given by Corollary 5.1. The column "field" names a field over which all factors are tight.

product	γ_{gr}	field
$P_k \boxtimes P_\ell$	$(k-1)(\ell-1)$	$\mathbb{R}$
$P_k \boxtimes C_\ell$	$(k-1)(\ell-2)$	$\mathbb{R}$
$C_k \boxtimes C_\ell$	$(k-2)(\ell-2)$	$\mathbb{C}$
$C_{k_1} \boxtimes \cdots \boxtimes C_{k_r}$	$\prod(k_i-2)$	$\mathbb{C}$
$C_k \boxtimes K_{a,b}$, $a \geq b$	$a(k-2)$	$\mathbb{C}$
$C_k \boxtimes CP(s)$	$2(k-2)$	$\mathbb{C}$
Petersen $\boxtimes C_k$	$5(k-2)$	$\mathbb{C}$
Petersen $\boxtimes$ Petersen	25	$\mathbb{C}$
$K(n,2) \boxtimes C_k$, n even, k odd	$(n-1)(k-2)$	$\mathbb{F}_{2^s}$
$K(n,2) \boxtimes K(m,2)$, n,m even	$(n-1)(m-1)$	$\mathbb{F}_2$
$K(7,2) \boxtimes K(10,2)$	54	$\mathbb{F}_3$
$K(9,2) \boxtimes C_k$, $5 \nmid k$	$8(k-2)$	$\mathbb{F}_{5^s}$

Corollary 5.3. *Let p be a prime and let $K(n_1,r_1),\dots,K(n_s,r_s)$ be Kneser graphs such that for each i some power $p^{a_i} \geq r_i$ of p divides $n_i - 2r_i$. Then*

$$\gamma_{\mathrm{gr}}\big(K(n_1,r_1) \boxtimes \cdots \boxtimes K(n_s,r_s)\big) = \prod_{i=1}^{s} \binom{n_i-1}{r_i-1},$$

and the same holds after adding complete graphs and cycles C_k with $p \nmid k$ as further factors, each cycle contributing $k-2$. For instance $\gamma_{\mathrm{gr}}(K(10,3) \boxtimes K(12,4)) = 36 \cdot 165 = 5940$, over $\mathbb{F}_2$.

Proof. The Kneser graphs are $\mathbb{F}_p$-tight by Theorem 4.4. If cycles are present, choose s such that $\mathbb{F}_{p^s}$ contains a primitive k-th root of unity for each cycle length k, which is possible because $p \nmid k$; then all factors are $\mathbb{F}_{p^s}$-tight by (2.1) and Proposition 3.6, whose last statement covers C_4 when p is odd, and Corollary 2.5 applies. □

Remark 5.4. For $3 \leq k \leq \ell \leq 7$ the value $\gamma_{\mathrm{gr}}(C_k \boxtimes C_\ell) = (k-2)(\ell-2)$ was also confirmed by an exhaustive computation of the longest dominating sequence (Section 9); for instance $\gamma_{\mathrm{gr}}(C_7 \boxtimes C_7) = 25$ on 49 vertices, and $\gamma_{\mathrm{gr}}(\text{Petersen} \boxtimes C_5) = 15$. The proofs do not use these computations.

5.1. **Strong products with a fixed tight factor.** Corollary 2.5 needs both factors tight. When only one factor is tight, the rank bound still gives an inequality.

Proposition 5.5. *If G is $\mathbb{F}$-tight, then for every graph H,*

$$\gamma_{\mathrm{gr}}(G)\,\gamma_{\mathrm{gr}}(H) \leq \gamma_{\mathrm{gr}}(G \boxtimes H) \leq \gamma_{\mathrm{gr}}(G)\ \mathrm{mr}_{\mathbb{F}}(H).$$

Proof. The lower bound is Lemma 2.3, and the upper bound is Theorem A with $\mathrm{mr}_{\mathbb{F}}(G) = \gamma_{\mathrm{gr}}(G)$. □

So for a tight G the gap in Conjecture 1.1 is controlled by the gap $\mathrm{mr}_{\mathbb{F}}(H) - \gamma_{\mathrm{gr}}(H)$ of the other factor; the octahedron is tight, and the graphs H_N of Theorem D have a large gap.

6. The strong product conjecture fails

By Corollary 2.5, a counterexample to Conjecture 1.1 must contain, for each characteristic, a factor that is not tight. The first construction uses two factors bounded over different fields, so

that no common field is available; the second uses a tight factor, the octahedron, and a factor that is tight over no field at all.

6.1. **Unbounded ratios.**

Theorem 6.1. *For $m \geq 5$ let A_m and B_m be the graphs on the five-element subsets of $[m]$ in which distinct sets S, T are adjacent when $|S \cap T|$ is odd, respectively when $|S \cap T| = 2$. Then*

$$\gamma_{\mathrm{gr}}(A_m) \leq m, \qquad \gamma_{\mathrm{gr}}(B_m) \leq \binom{m}{2}, \qquad \gamma_{\mathrm{gr}}(A_m \boxtimes B_m) \geq \binom{m}{5}.$$

Consequently

$$\frac{\gamma_{\mathrm{gr}}(A_m \boxtimes B_m)}{\gamma_{\mathrm{gr}}(A_m)\,\gamma_{\mathrm{gr}}(B_m)} \;\geq\; \frac{(m-2)(m-3)(m-4)}{60\,m} \;\longrightarrow\; \infty.$$

Proof. Let U be the incidence matrix between the five-element subsets of $[m]$ and the elements of $[m]$. Over $\mathbb{F}_2$ the matrix UU^{T} has entries $|S \cap T| \bmod 2$. The diagonal entries are $5 \equiv 1$, and an off-diagonal entry is nonzero exactly when $|S \cap T|$ is odd. So $UU^{\mathsf{T}} \in \mathcal{S}_{\mathbb{F}_2}[A_m]$, its rank is at most m, and Lemma 2.1 gives $\gamma_{\mathrm{gr}}(A_m) \leq m$.

Let V be the incidence matrix between five-element subsets and two-element subsets of $[m]$, with $V_{S,P} = 1$ when $P \subseteq S$. Over $\mathbb{F}_3$, $(VV^{\mathsf{T}})_{ST} = \binom{|S\cap T|}{2} \bmod 3$, the number of pairs inside $S \cap T$. For $t = 0, 1, \ldots, 5$ the binomial coefficients $\binom{t}{2} = 0, 0, 1, 3, 6, 10$ are congruent to $0, 0, 1, 0, 0, 1$ modulo 3, so the nonzero entries are those with $S = T$ or $|S \cap T| = 2$. Hence $VV^{\mathsf{T}} \in \mathcal{S}_{\mathbb{F}_3}[B_m]$ has rank at most $\binom{m}{2}$, and $\gamma_{\mathrm{gr}}(B_m) \leq \binom{m}{2}$.

Finally, the diagonal $\{(S, S)\}$ is an independent set of $A_m \boxtimes B_m$: for $S \neq T$ adjacency of (S, S) and (T, T) would require $|S \cap T|$ to be odd and equal to 2. Listing an independent set in any order gives a dominating sequence, each vertex footprinting itself. This gives the lower bound, and the ratio is

$$\binom{m}{5} \Big/ \Big(m\binom{m}{2}\Big) = \frac{(m-2)(m-3)(m-4)}{60m}. \qquad \square$$

For $m = 13$ the product has a dominating sequence of length 1287, while $\gamma_{\mathrm{gr}}(A_{13})\gamma_{\mathrm{gr}}(B_{13}) \leq 13 \cdot 78 = 1014$. The two factor bounds are over $\mathbb{F}_2$ and $\mathbb{F}_3$. Over a common field the product of the two matrices would bound the product graph, by Theorem A, so the change of field is forced. The construction is in the spirit of the intersection-parity arguments of Alon for the Shannon capacity of a disjoint union [1], with the difference that Lemma 2.1 needs the nonzero entries to be exactly the closed neighbourhoods, not merely the diagonal.[6]

6.2. **The optimal order of magnitude.**

In Theorem 6.1 the ratio grows like $n^{2/5}$, where $n = \binom{m}{5}$ is the number of vertices of each factor. Replacing the primes 2 and 3 by larger primes raises the exponent, and a simple projection argument shows that the exponent cannot exceed $1/2$ when the ratio is measured against the order of the product. The construction is a two-prime version of the modular intersection graphs of Frankl and Wilson [16].

Fix distinct primes p and q, put $k = pq - 1$, and let $m \geq k$. On the vertex set $\binom{[m]}{k}$ define graphs $G_p(m)$ and $G_q(m)$ by

$$S \sim_{G_p} T \iff |S \cap T| \equiv -1 \pmod{p}, \quad S \sim_{G_q} T \iff |S \cap T| \equiv -1 \pmod{q} \quad (S \neq T).$$

[6]For an independent set the diagonal alone would suffice. The difference between bounding the independence number and bounding the Grundy domination number is that the off-diagonal support must match the edges.

Lemma 6.2. *$E(G_p(m)) \cap E(G_q(m)) = \emptyset$ and*

$$\gamma_{\mathrm{gr}}(G_p(m)) \le R_p(m) := \sum_{i=0}^{p-1} \binom{m}{i}, \qquad \gamma_{\mathrm{gr}}(G_q(m)) \le R_q(m).$$

Proof. If $S \neq T$ then $0 \le |S \cap T| \le k-1 = pq-2$, and $|S \cap T| \equiv -1$ modulo both p and q would force $|S \cap T| \equiv -1 \pmod{pq}$, which is impossible in that range. For the rank bound let $f_p(t) = 1-(t+1)^{p-1} \in \mathbb{Z}[t]$. By Fermat's little theorem $f_p(t) \equiv 1 \pmod p$ if $t \equiv -1$ and $f_p(t) \equiv 0$ otherwise. Since $|S \cap S| = k \equiv -1 \pmod p$, the matrix $\big(f_p(|S \cap T|) \bmod p\big)_{S,T}$ is a closed pattern matrix of $G_p(m)$, and by (4.3) its rank is at most $R_p(m)$, because $\deg f_p = p-1$. Apply Lemma 2.1. The same argument works for q. □

For $(p,q) = (2,3)$ we have $k = 5$, the graphs $G_2(m)$ and $G_3(m)$ are the graphs A_m and B_m of Theorem 6.1, and the bounds improve: $f_2(t) = -t \equiv \binom{t}{1} \pmod 2$ and $f_3(t) = -t^2 - 2t = -2\binom{t}{2} - 3t \equiv \binom{t}{2} \pmod 3$, so by (4.2) the matrices are $W_1^{\mathsf T} W_1$ over $\mathbb{F}_2$ and $W_2^{\mathsf T} W_2$ over $\mathbb{F}_3$, which are the matrices $UU^{\mathsf T}$ and $VV^{\mathsf T}$ of that proof, of ranks at most m and $\binom{m}{2}$. For larger primes the rank bound of Lemma 6.2 is what we use.

Theorem 6.3. *Let $p < q$ be primes, $k = pq-1$ and $\theta_{p,q} = (p-1)(q-1)/(pq-1)$. For every $N \ge 2$ there is a graph G and a graph H, both with N vertices, such that*

$$\frac{\gamma_{\mathrm{gr}}(G \boxtimes H)}{\gamma_{\mathrm{gr}}(G)\,\gamma_{\mathrm{gr}}(H)} \;\ge\; \frac{N^{\theta_{p,q}}}{(k+2)^{p+q-2}}.$$

Proof. Let m be the least integer with $\binom{m}{k} \ge N$; since $N \ge 2$, $m \ge k+1$. Choose a set W of N k-subsets of $[m]$ and put $G = G_p(m)[W]$, $H = G_q(m)[W]$. Since G and H have no common edge (Lemma 6.2), the diagonal $\{(S,S) : S \in W\}$ is an independent set of $G \boxtimes H$, as in the proof of Theorem 6.1, so $\gamma_{\mathrm{gr}}(G \boxtimes H) \ge N$; and by Lemmas 3.1 and 6.2, $\gamma_{\mathrm{gr}}(G) \le R_p(m)$ and $\gamma_{\mathrm{gr}}(H) \le R_q(m)$. Since $\binom{m}{i} \le m^i \le \binom{p-1}{i} m^i$ for $i \le p-1$, we have $R_p(m) \le (m+1)^{p-1}$, and similarly $R_q(m) \le (m+1)^{q-1}$. By the choice of m and the bound $\binom{a}{k} \ge (a/k)^k$ for $a \ge k$, we get $((m-1)/k)^k \le \binom{m-1}{k} < N$, so $m+1 < kN^{1/k} + 2 \le (k+2)N^{1/k}$. Therefore

$$\frac{\gamma_{\mathrm{gr}}(G \boxtimes H)}{\gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H)} \ge \frac{N}{(m+1)^{p+q-2}} > \frac{N^{1-(p+q-2)/k}}{(k+2)^{p+q-2}},$$

and $1 - \frac{p+q-2}{pq-1} = \frac{(p-1)(q-1)}{pq-1} = \theta_{p,q}$. □

The exponent $\theta_{p,q}$ tends to 1 as $p, q \to \infty$; for $(p,q) = (2,3)$ it is $2/5$, for $(2,5)$ it is $4/9$ and for $(3,5)$ it is $4/7$. The next proposition shows that, measured in the order of the product, nothing better than exponent $1/2$ is possible, so Theorem 6.3 is optimal up to the $o(1)$ in the exponent.

Proposition 6.4. *For all graphs G and H,*

$$\gamma_{\mathrm{gr}}(G \boxtimes H) \le \min\{|V(G)|\,\gamma_{\mathrm{gr}}(H),\ \gamma_{\mathrm{gr}}(G)\,|V(H)|\}.$$

Consequently $\gamma_{\mathrm{gr}}(G \boxtimes H)/(\gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H)) \le \min\{|V(G)|, |V(H)|\} \le |V(G \boxtimes H)|^{1/2}$.

Proof. Let $((g_1,h_1),\dots,(g_\ell,h_\ell))$ be a dominating sequence of $G \boxtimes H$ with footprints $(a_1,b_1),\dots,(a_\ell,b_\ell)$. For a vertex a of G let $I_a = \{i : a_i = a\}$. If $j < i$ are in I_a, then $(a,b_i) \notin N[g_j] \times N[h_j]$ while $a = a_j \in N[g_j]$; hence $b_i \notin N[h_j]$. Also $b_i \in N[h_i]$. So the subsequence $(h_i)_{i \in I_a}$ is a dominating sequence of H with footprints b_i, and $|I_a| \le \gamma_{\mathrm{gr}}(H)$. Summing over $a \in V(G)$ gives $\ell \le |V(G)|\gamma_{\mathrm{gr}}(H)$; the other bound is symmetric. The ratio bound follows because $\gamma_{\mathrm{gr}} \ge 1$. □

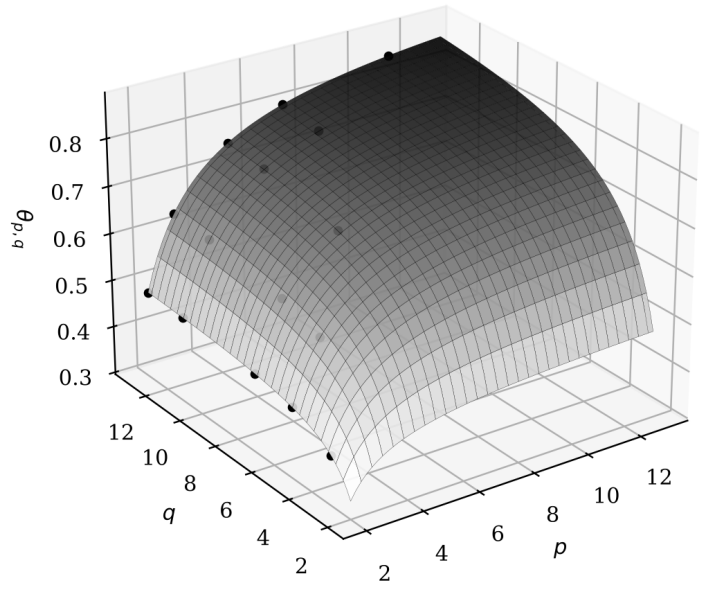


FIGURE 5. The exponent $\theta_{p,q} = (p-1)(q-1)/(pq-1)$ of Theorem 6.3 for pairs of primes $p < q$. It tends to 1, which corresponds to the exponent $1/2$ in the order of the product; Proposition 6.4 shows that $1/2$ cannot be exceeded.

Corollary 6.5. *Let $\rho(n)$ be the largest value of $\gamma_{\mathrm{gr}}(G \boxtimes H)/(\gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H))$ over graphs G, H with $|V(G)||V(H)| \le n$. Then $\rho(n) \le n^{1/2}$, and for every $\varepsilon > 0$ there is $c_\varepsilon > 0$ with $\rho(n) \ge c_\varepsilon n^{1/2-\varepsilon}$ for all $n \ge 4$. In particular the ratio grows as $n^{1/2+o(1)}$.*

Proof. The upper bound is Proposition 6.4. Given ε, choose primes $p < q$ with $\theta_{p,q} > 1 - 2\varepsilon$, and apply Theorem 6.3 with $N = \lfloor n^{1/2} \rfloor \ge 2$; then $N \ge n^{1/2}/2$ and the ratio is at least $(n^{1/2}/2)^{1-2\varepsilon}/(pq+1)^{p+q-2}$. □

6.3. **The octahedron.** The octahedron is the cocktail party graph $K_{2,2,2} = CP(3)$. Write its vertices as $g_0, g_1, g_2, x_0, x_1, x_2$, with x_t the antipode of g_t, and read indices modulo 3. By Proposition 3.5, $\gamma_{\mathrm{gr}}(K_{2,2,2}) = 2$.

Proposition 6.6. *Let H be a graph with vertices $v_0, \dots, v_{N-1}$ and let $t : V(H) \to \mathbb{Z}/3$ satisfy $t(v_i) = t(v_j) + 1$ for every edge $v_i v_j$ with $i < j$. Then*

$$s_j = (g_{t(v_j)}, v_j), \qquad j = 0, 1, \dots, N-1,$$

is a dominating sequence of $K_{2,2,2} \boxtimes H$, the vertex s_j footprinting $f_j = (x_{t(v_j)+1}, v_j)$. Hence $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H) \ge N$.

Proof. The vertex $x_{t(v_j)+1}$ is not the antipode $x_{t(v_j)}$ of $g_{t(v_j)}$, so it lies in $N[g_{t(v_j)}]$, and $f_j \in N[s_j]$. Let $i < j$ and suppose $f_j \in N[s_i]$. Then $x_{t(v_j)+1} \in N[g_{t(v_i)}]$, so $t(v_i) \ne t(v_j) + 1$; and $v_j \in N[v_i]$ with $v_i \ne v_j$, so $v_i v_j$ is an edge and $t(v_i) = t(v_j) + 1$. This is a contradiction. □

The colouring rotates the three layers x_0, x_1, x_2 of the octahedron along the sequence, so that every new vertex reaches a footprint in a layer that the earlier vertices of its column have left uncovered (Figure 6). For H_N, the graph on $\{0, \dots, N-1\}$ with $u \sim v$ when $|u - v| \equiv 1$

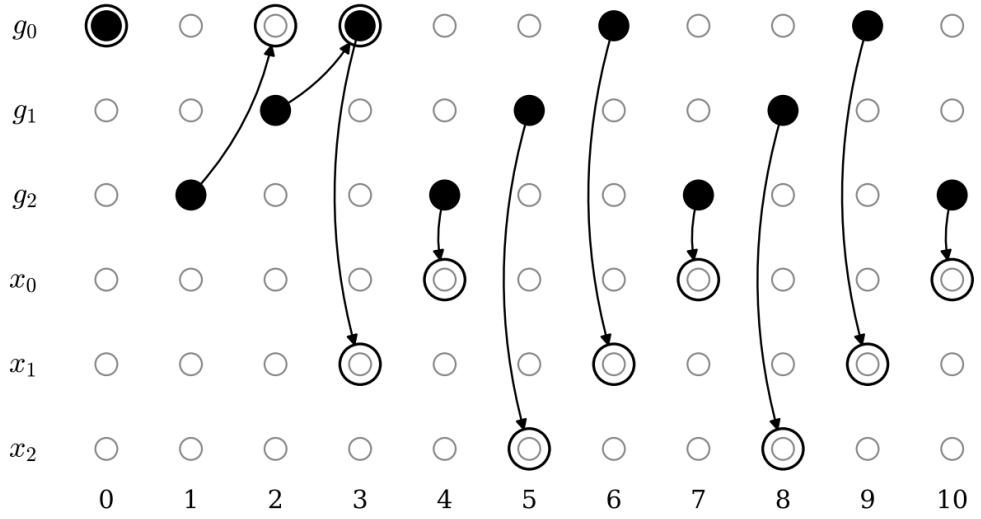


FIGURE 6. The strong product $K_{2,2,2} \boxtimes H_{11}$ drawn as six layers, one for each vertex of the octahedron, over the 11 vertices of H_{11}. The dominating sequence of Proposition 6.6 uses one vertex of each column, in the layers $g_0, g_2, g_1, g_0, \ldots$, and each circled vertex is the footprint of the term joined to it by an arrow (the first term is its own footprint).

(mod 3), the colouring $t(v) = -v \bmod 3$ has the required property: if $u < v$ are adjacent, then $v - u \equiv 1$, so $t(u) \equiv t(v) + 1$. Therefore

$$\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N) \geq N \qquad (N \geq 1). \tag{6.1}$$

6.4. **The Grundy domination number of H_N.** Write $a_i = 3i$, $b_i = 3i + 1$ and $c_i = 3i + 2$ for the vertices of H_N. Two vertices of one residue class are never adjacent, and comparing differences modulo 3 gives

$$a_i \sim b_j \iff j \geq i, \qquad b_i \sim c_j \iff j \geq i, \qquad c_i \sim a_j \iff j > i. \tag{6.2}$$

For example $b_j - a_i = 3(j - i) + 1$, which is congruent to 1 when $j \geq i$ and has absolute value $3(i - j) - 1 \equiv 2$ when $j < i$; the other two cases are the same computation.

Lemma 6.7. *For $N \geq 4$, $\gamma_{\mathrm{gr}}(H_N) \geq \lfloor N/3 \rfloor + 2$.*

Proof. Let $q = \lfloor N/3 \rfloor$. By Lemma 3.1 it suffices to find a dominating sequence of length $q + 2$ in an induced subgraph $H_{N'}$ with $N' \leq N$. For $q = 1$, H_4 is the path $0, 1, 2, 3$ and $(0, 1, 2)$ has the footprints $0, 2, 3$. For $q = 2$, the sequence $(0, 5, 1, 2)$ of H_6 has the footprints $0, 5, 2, 3$. For $q \geq 3$ consider in H_{3q} the sequence

$$a_0,\ a_1,\ c_0,\ c_1,\ \ldots,\ c_{q-2},\ b_0,$$

of length $q + 2$. By (6.2): a_0 footprints itself; a_1 footprints itself, since it is not adjacent to a_0; c_0 is adjacent to a_2, which neither a_0 nor a_1 dominates, so a_2 is a footprint; for $1 \leq m \leq q - 2$ the vertex c_m is adjacent to no earlier vertex (the only earlier vertex adjacent to a c is a_1, adjacent to c_0 alone), so it footprints itself; and b_0 is adjacent to c_{q-1}, which no earlier vertex dominates. □

The upper bound comes from the structure of the vertices not dominated by the first vertex of a sequence.

Lemma 6.8. *Let $q \geq 1$ and let z be a vertex of a graph Γ whose non-neighbours are $2q$ vertices $X_0, \ldots, X_{q-1}$, $Y_0, \ldots, Y_{q-1}$, with no edges among the X_i, none among the Y_j, and $X_i \sim Y_j$ exactly when $i \geq j$. Suppose that every neighbour w of z has an index $k \in \{0, \ldots, q\}$ with*

$$N[w] \cap \{X_i, Y_j\} = \{X_i : i < k\} \cup \{Y_j : j \geq k\}.$$

Then every dominating sequence of Γ beginning with z has length at most $q + 2$.

Proof. Let $(z, v_1, \ldots, v_m)$ be a dominating sequence. After z, the undominated vertices are the X_i and Y_j, which induce a half graph containing the perfect matching $\{X_iY_i\}$. We use the following counting principle: if after some step exactly u vertices are undominated, at most u further steps follow, since each step dominates a new vertex.

Let τ be the first index for which v_τ is a neighbour of z, or v_τ is an X or a Y adjacent to an earlier v_i. If there is no such index, the v_i form an independent set of the half graph, which contains at most one end of each matching edge, so $m \leq q$. Otherwise let I and J be the sets of indices of the X's and Y's among $v_1, \ldots, v_{\tau-1}$, with $r = |I|$ and $s = |J|$, so $\tau - 1 = r + s$. Since no two of them are adjacent, $i < j$ for all $i \in I$ and $j \in J$.

Case 1: $v_\tau = X_i$ is adjacent to an earlier Y_j. Then $J \neq \emptyset$ and $b = \min J \leq i$, and all indices in I are less than b. After step τ an X-vertex is dominated if it was selected or is adjacent to a selected Y, that is, if its index lies in $I \cup \{i\}$ or is at least b; the undominated X-vertices are the $b - r$ vertices with index in $\{0, \ldots, b-1\} \setminus I$. A Y-vertex is dominated if it was selected or is adjacent to a selected X, and the largest selected X-index is i; with $s_0 = |J \cap \{b, \ldots, i\}|$, the undominated Y-vertices number $(q - 1 - i) - (s - s_0)$. By the counting principle,

$$m \leq \tau + (b - r) + (q - 1 - i) - (s - s_0) = q + b - i + s_0 \leq q + 1,$$

because $s_0 \leq i - b + 1$.

Case 2: $v_\tau = Y_j$ is adjacent to an earlier X_i. The map $X_i \mapsto Y_{q-1-i}$, $Y_j \mapsto X_{q-1-j}$ preserves the half graph, since $q - 1 - j \geq q - 1 - i$ exactly when $i \geq j$, and before step τ only X's and Y's were selected; the count of Case 1 applies verbatim and gives $m \leq q + 1$.

Case 3: $v_\tau = w$ is a neighbour of z with index k. After step τ every undominated vertex lies in $W_k = \{X_i : i \geq k\} \cup \{Y_j : j < k\}$, a set of q vertices. We claim that at least $r + s$ vertices of W_k were dominated before step τ. Send a selected X_i to X_i if $i \geq k$ and to Y_i if $i < k$, and a selected Y_j to Y_j if $j < k$ and to X_j if $j \geq k$. Each image lies in W_k and is dominated by the selected vertex, which is either the image itself or its partner in the matching. Distinct selections on one side have distinct images, and a selection X_i and a selection Y_j could share an image only if $i = j$, which is impossible because $i < j$. So the map is injective, at most $q - r - s$ vertices remain undominated after step τ, and

$$m \leq \tau + q - r - s = q + 1.$$

In all cases the sequence $(z, v_1, \ldots, v_m)$ has length at most $q + 2$. □

Lemma 6.9. *For every $q \geq 1$, $\gamma_{\mathrm{gr}}(H_{3q+2}) \leq q + 2$.*

Proof. Let $n = 3q + 2$. For $1 \leq d < n$ we have $d \equiv 1 \pmod 3$ if and only if $n - d \equiv 1 \pmod 3$, because $n \equiv 2$. Hence $u \sim v$ in H_n exactly when $(u - v) \bmod n \equiv 1 \pmod 3$, and translation modulo n is an automorphism. A dominating sequence remains one under an automorphism, so we may assume that it begins with $z = 0$. The neighbours of 0 are $b_0, \ldots, b_q$. Its non-neighbours are $X_i = a_{i+1}$ $(0 \leq i < q)$ and $Y_j = c_j$ $(0 \leq j < q)$. By (6.2), $X_i = a_{i+1} \sim c_j$ exactly when $i + 1 > j$, that is, $i \geq j$; and b_k is adjacent to a_{i+1} exactly when $i < k$ and to c_j exactly when $j \geq k$. Lemma 6.8 applies. □

Theorem 6.10. *For $N \geq 4$, $\gamma_{\mathrm{gr}}(H_N) = \lfloor N/3 \rfloor + 2$ and $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N) \geq N$. Conjecture 1.1 fails for $K_{2,2,2}$ and H_N exactly when $N > 2\gamma_{\mathrm{gr}}(H_N)$, that is, for $N \in \{11, 13, 14\}$ and all $N \geq 15$.*

Proof. With $q = \lfloor N/3 \rfloor$, H_N is an induced subgraph of H_{3q+2}, so Lemmas 6.7, 3.1 and 6.9 give $\gamma_{\mathrm{gr}}(H_N) = q + 2$. The lower bound for the product is (6.1). Write $N = 3q + e$ with $e \in \{0, 1, 2\}$; then $N - 2(q+2) = q + e - 4 > 0$ exactly when $q \geq 5 - e$, which gives the stated values.[7] □

The smallest of these counterexamples is $K_{2,2,2} \boxtimes H_{11}$, with 66 vertices: $\gamma_{\mathrm{gr}}(K_{2,2,2}) = 2$ and $\gamma_{\mathrm{gr}}(H_{11}) = 5$, while the product has the dominating sequence $(g_{-j \bmod 3}, j)$, $0 \leq j \leq 10$, of length 11.

Corollary 6.11. *For $N \in \{11, 13, 14\}$ and $N \geq 15$, the graph H_N is not tight over any field: $\gamma_{\mathrm{gr}}(H_N) < \mathrm{mr}_{\mathbb{F}}(H_N)$ for every field $\mathbb{F}$.*

Proof. Suppose H_N were $\mathbb{F}$-tight. Let $\mathbb{F}' \supseteq \mathbb{F}$ be a finite extension with at least five elements. By (2.1), H_N is $\mathbb{F}'$-tight, and $K_{2,2,2}$ is $\mathbb{F}'$-tight by Proposition 3.5. Corollary 2.5 would give $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N) = 2\gamma_{\mathrm{gr}}(H_N) < N$, contradicting (6.1). □

So the octahedron, although it is the factor with the small Grundy number, is not the source of the failure; the graphs H_N are. In the language of Proposition 5.5, for every field $\mathbb{F}$ and every N as above,

$$\mathrm{mr}_{\mathbb{F}}(H_N) \;\geq\; \tfrac{1}{2}\,\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N) \;\geq\; \tfrac{N}{2},$$

which is roughly $\frac{3}{2}\gamma_{\mathrm{gr}}(H_N)$. With the octahedron fixed, the ratio $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N)/(2\gamma_{\mathrm{gr}}(H_N))$ tends to $3/2$; unbounded ratios require both factors to vary, as in Theorem 6.1.

7. Direct products and Grundy total domination

Throughout this section all graphs are without isolated vertices, so that open neighbourhoods are nonempty and γ^t_{gr} is defined. A vertex u is a *total footprint* of v_i in a sequence $(v_1, \ldots, v_k)$ if $u \in N(v_i) \setminus \bigcup_{j<i} N(v_j)$, and the sequence is a total dominating sequence if every term has a total footprint.

7.1. **Totally tight graphs.**

Lemma 7.1. *If H is an induced subgraph of G and neither graph has isolated vertices, then $\gamma^t_{\mathrm{gr}}(H) \leq \gamma^t_{\mathrm{gr}}(G)$. The Grundy total domination number of a disjoint union is the sum of those of the components.*

Proof. Let $(v_1, \ldots, v_k)$ be a total dominating sequence of H with total footprints $u_1, \ldots, u_k$. For vertices u, v of H, $u \in N_G(v)$ if and only if $u \in N_H(v)$, since H is induced. Hence $u_i \in N_G(v_i)$ and $u_i \notin N_G(v_j)$ for $j < i$, and the same sequence is a total dominating sequence of G. For a disjoint union, the subsequence of a total dominating sequence lying in one component is a total dominating sequence of that component, and conversely the concatenation of sequences of the components is one of the union. □

Recall that the matching number $\nu(F)$ of a graph F is the largest number of pairwise disjoint edges.

Proposition 7.2. *Let F be a forest without isolated vertices. Then $\gamma^t_{\mathrm{gr}}(F) = 2\nu(F)$, and F is totally $\mathbb{F}$-tight over every field $\mathbb{F}$: its adjacency matrix has rank $2\nu(F)$ over $\mathbb{F}$.*

[7] "Exactly" refers to the inequality $N > 2\gamma_{\mathrm{gr}}(H_N)$; for $N = 12$ the lower bound (6.1) equals $2\gamma_{\mathrm{gr}}(H_{12})$ and gives no counterexample, and the exact value $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_{12}) = 12$ found by computation (Section 9) shows that there is none.

Proof. We prove $\gamma^t_{\mathrm{gr}}(F) \geq 2\nu(F)$ by induction on $|V(F)|$. By Lemma 7.1 it suffices to treat a tree T. If $T = K_2$ the claim is $\gamma^t_{\mathrm{gr}}(K_2) = 2$, which is clear. Otherwise let $\ell s \cdots$ be a longest path of T, beginning at the leaf ℓ. Every neighbour of s other than its successor p on the path is a leaf, for otherwise the path could be lengthened. Let L be the set of leaves adjacent to s and $T' = T - (\{s\} \cup L)$, with the vertex p removed as well if it becomes isolated. Every matching of T contains at most one edge meeting $\{s\} \cup L$, so $\nu(T) \leq \nu(T') + 1$. Let $(x_1, \ldots, x_m)$ be a total dominating sequence of T' of length $\gamma^t_{\mathrm{gr}}(T')$ (empty if T' is empty). We claim that

$$(\ell,\ x_1, \ldots, x_m,\ s)$$

is a total dominating sequence of T. The first term has total footprint s. A total footprint $y_i \in V(T')$ of x_i in T' is not adjacent to ℓ, because $N(\ell) = \{s\}$, and is not adjacent in T to any earlier x_j, because T' is induced. The last term has total footprint ℓ, since s is the only neighbour of ℓ. By induction $\gamma^t_{\mathrm{gr}}(T) \geq 2 + 2\nu(T') \geq 2\nu(T)$.

Conversely, let A be the adjacency matrix of F, a symmetric matrix with zero diagonal, so $A \in \mathcal{S}_{\mathbb{F}}(F)$ for every field $\mathbb{F}$. The rank of a symmetric matrix over any field equals the order of its largest nonsingular principal submatrix. A principal submatrix of A is the adjacency matrix of an induced subforest $F[S]$, and its determinant is $\pm$ the number of perfect matchings of $F[S]$, because a permutation contributes to the determinant only if all of its cycles have length two when the graph has no cycles. A forest has at most one perfect matching, so $\det A[S] = \pm 1$ if $F[S]$ has a perfect matching and 0 otherwise, in every characteristic. The largest such S is the vertex set of a maximum matching, so $\operatorname{rank}_{\mathbb{F}} A = 2\nu(F)$. With Lemma 2.1, $2\nu(F) \leq \gamma^t_{\mathrm{gr}}(F) \leq \mathrm{mr}^{\circ}_{\mathbb{F}}(F) \leq 2\nu(F)$. □

By König's theorem $\nu(F)$ is also the least size $\beta(F)$ of a vertex cover of F, and the value $\gamma^t_{\mathrm{gr}}(T) = 2\beta(T)$ for trees is due to Brešar, Kos, Nasini and Torres [12]; the proof above is included because the rank statement is what Theorem 7.4 needs. For the path, $\gamma^t_{\mathrm{gr}}(P_m) = 2\lfloor m/2 \rfloor$.

Proposition 7.3. (i) *For $a, b \geq 1$, $\gamma^t_{\mathrm{gr}}(K_{a,b}) = \mathrm{mr}^{\circ}_{\mathbb{F}}(K_{a,b}) = 2$ over every field.*

(ii) *For $n \geq 2$, $\gamma^t_{\mathrm{gr}}(K_n) = 2$, and K_n is totally $\mathbb{F}$-tight over every field with at least n elements.*

(iii) *For $n \geq 3$, $\gamma^t_{\mathrm{gr}}(C_n) = 2\lfloor (n-1)/2 \rfloor$, that is, $n-1$ for odd n and $n-2$ for even n. The cycle C_n is totally $\mathbb{R}$-tight for even n and totally $\mathbb{C}$-tight for every n.*

Proof. (i) The matrix $\left(\begin{smallmatrix} 0 & J \\ J & 0 \end{smallmatrix}\right)$ lies in $\mathcal{S}_{\mathbb{F}}(K_{a,b})$ and has rank 2. A vertex of one side and then a vertex of the other side form a total dominating sequence, and nothing can follow, since together they dominate every vertex.

(ii) Choose distinct $x_v \in \mathbb{F}$ and set $M_{uv} = x_u - x_v$. The diagonal vanishes and the off-diagonal entries do not, and $M = x\mathbf{1}^{\mathsf{T}} - \mathbf{1}x^{\mathsf{T}}$ has rank at most 2. The sequence (v, w) with $v \neq w$ has total footprints w and v.

(iii) Deleting a vertex of C_n leaves an induced path P_{n-1}, so Lemma 7.1 and Proposition 7.2 give $\gamma^t_{\mathrm{gr}}(C_n) \geq 2\lfloor (n-1)/2 \rfloor$. For the upper bound let $\omega = e^{i\theta}$ and let A_ω be the Hermitian matrix with $(A_\omega)_{t,t+1} = 1$ and $(A_\omega)_{t+1,t} = 1$ for $0 \leq t \leq n-2$, $(A_\omega)_{n-1,0} = \omega$ and $(A_\omega)_{0,n-1} = \bar{\omega}$. It lies in $\mathcal{S}_{\mathbb{C}}(C_n)$. With $\alpha = e^{i\theta/n}$ and $D = \operatorname{diag}(\alpha^{-t})_{0 \leq t < n}$, one checks entry by entry that $DA_\omega D^{-1} = \alpha S + \bar{\alpha} S^{-1}$, where S is the permutation matrix with $S_{t,t+1} = 1$, indices modulo n; the eigenvalues of S are the n-th roots of unity, so the eigenvalues of A_ω are

$$2\cos\frac{\theta + 2\pi j}{n}, \qquad 0 \leq j \leq n-1.$$

If $n \equiv 0 \pmod 4$ take $\theta = 0$, and if $n \equiv 2 \pmod 4$ take $\theta = \pi$; in both cases exactly two of the angles $(\theta + 2\pi j)/n$ are odd multiples of $\pi/2$, and A_ω is a real matrix of rank $n-2$. If n is odd

TABLE 4. Values of γ_{gr}^t for direct products given by Theorem 7.4 and confirmed by the exact solver of Appendix C. Here T_7 is the complete binary tree with seven vertices, for which $\nu(T_7) = 2$.

product	vertices	γ_{gr}^t	product	vertices	γ_{gr}^t
$C_3 \times C_3$	9	4	$C_6 \times C_5$	30	16
$C_3 \times C_5$	15	8	$C_5 \times C_7$	35	24
$C_4 \times C_5$	20	8	$C_3 \times C_3 \times C_3$	27	8
$C_5 \times C_5$	25	16	$K_4 \times C_5$	20	8
$C_3 \times C_7$	21	12	$T_7 \times C_5$	35	16

take $\theta = \pi/2$; then $(\pi/2 + 2\pi j)/n \equiv \pi/2 \pmod{\pi}$ means $4j + 1 \equiv n \pmod{2n}$ or $4j + 1 \equiv 3n \pmod{2n}$, and exactly one $j \in \{0, \dots, n-1\}$ satisfies one of these, so A_ω has rank $n - 1$. □

Theorem 7.4. *Let $\mathbb{F}$ be a field. Direct products of totally $\mathbb{F}$-tight graphs are totally $\mathbb{F}$-tight, and for totally $\mathbb{F}$-tight graphs $G_1, \dots, G_r$,*

$$\gamma_{\mathrm{gr}}^t(G_1 \times \cdots \times G_r) = \prod_{i=1}^{r} \gamma_{\mathrm{gr}}^t(G_i).$$

In particular, for every $r \geq 1$, all $k_1, \dots, k_r \geq 3$ and all forests $F_1, \dots, F_s$ without isolated vertices,

$$\gamma_{\mathrm{gr}}^t(C_{k_1} \times \cdots \times C_{k_r} \times F_1 \times \cdots \times F_s) = \prod_{i=1}^{r} 2\Big\lfloor \frac{k_i - 1}{2} \Big\rfloor \prod_{j=1}^{s} 2\nu(F_j),$$

and factors K_n and $K_{a,b}$ may be added, each contributing a factor 2.

Proof. If $M \in \mathcal{S}_{\mathbb{F}}(G)$ and $M' \in \mathcal{S}_{\mathbb{F}}(H)$, then $M \otimes M'$ has its nonzero entries exactly at the pairs $((g,h),(g',h'))$ with $g' \in N(g)$ and $h' \in N(h)$, so $M \otimes M' \in \mathcal{S}_{\mathbb{F}}(G \times H)$ and $\mathrm{mr}_{\mathbb{F}}^{\circ}(G \times H) \leq \mathrm{mr}_{\mathbb{F}}^{\circ}(G)\,\mathrm{mr}_{\mathbb{F}}^{\circ}(H)$. Together with (1.3) and Lemma 2.1,

$$\gamma_{\mathrm{gr}}^t(G)\gamma_{\mathrm{gr}}^t(H) \leq \gamma_{\mathrm{gr}}^t(G \times H) \leq \mathrm{mr}_{\mathbb{F}}^{\circ}(G \times H) \leq \mathrm{mr}_{\mathbb{F}}^{\circ}(G)\,\mathrm{mr}_{\mathbb{F}}^{\circ}(H) = \gamma_{\mathrm{gr}}^t(G)\gamma_{\mathrm{gr}}^t(H)$$

for totally $\mathbb{F}$-tight G and H, and equality holds throughout. Induction on r gives the general case. By Propositions 7.2 and 7.3 all cycles, forests without isolated vertices, complete graphs and complete bipartite graphs are totally $\mathbb{C}$-tight, which gives the displayed formula. □

For two cycles Theorem 7.4 recovers the theorem of [7]; their theorem for a tree factor allows an arbitrary second factor and is not a consequence of ours. Table 4 lists values confirmed by exact computation.

7.2. **From closed to open neighbourhoods.** For a graph G let $\mathsf{B}(G)$ be the bipartite graph with vertex set $V(G) \times \{0,1\}$ in which (u,a) and (v,b) are adjacent when $a \neq b$ and $v \in N[u]$. Since $u \in N[u]$, the graph $\mathsf{B}(G)$ has no isolated vertices; it is the direct product $G^{\circ} \times K_2$, where G° is G with a loop at every vertex (Figure 7).

Proposition 7.5. *For all graphs G and H,*

(i) $\gamma_{\mathrm{gr}}^t(\mathsf{B}(G)) = 2\gamma_{\mathrm{gr}}(G)$;
(ii) $\mathsf{B}(G) \times \mathsf{B}(H)$ *is isomorphic to the disjoint union of two copies of* $\mathsf{B}(G \boxtimes H)$, *so that* $\gamma_{\mathrm{gr}}^t(\mathsf{B}(G) \times \mathsf{B}(H)) = 4\gamma_{\mathrm{gr}}(G \boxtimes H)$.

Proof. (i) Let $(v_1, \dots, v_k)$ be a dominating sequence of G with footprints $u_1, \dots, u_k$. Then

$$\big((v_1,0), \dots, (v_k,0), (u_k,1), (u_{k-1},1), \dots, (u_1,1)\big)$$

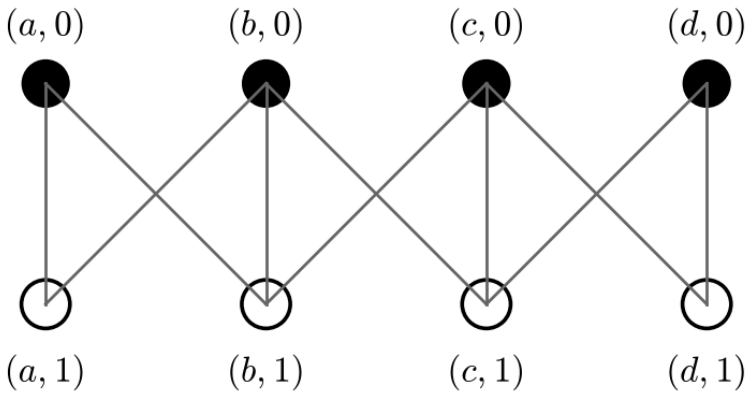


FIGURE 7. The bipartite double $\mathsf{B}(P_4)$ of the path $a\,b\,c\,d$: each vertex on the top row is joined to its closed neighbourhood on the bottom row. A dominating sequence of P_4, read forwards on the top row and its footprints read backwards on the bottom row, is a total dominating sequence of $\mathsf{B}(P_4)$ of twice the length.

is a total dominating sequence of $\mathsf{B}(G)$. Indeed $(u_i, 1)$ is a total footprint of $(v_i, 0)$, because the neighbours of $(v_j, 0)$ are the vertices $(w, 1)$ with $w \in N[v_j]$, and $u_i \notin N[v_j]$ for $j < i$. The vertex $(v_i, 0)$ is a total footprint of $(u_i, 1)$, because $v_i \in N[u_i]$, the earlier terms on side 0 dominate only side 1, and the earlier terms on side 1 are the $(u_j, 1)$ with $j > i$, which are not adjacent to $(v_i, 0)$ because $u_j \notin N[v_i]$. So $\gamma^t_{\mathrm{gr}}(\mathsf{B}(G)) \geq 2\gamma_{\mathrm{gr}}(G)$.

Conversely let σ be a total dominating sequence of $\mathsf{B}(G)$. Terms on side 0 dominate only vertices of side 1 and vice versa. So the terms $(x_1, 0), \ldots, (x_s, 0)$ of σ on side 0, in their order in σ, have total footprints $(y_1, 1), \ldots, (y_s, 1)$ with $y_i \in N[x_i] \setminus \bigcup_{j<i} N[x_j]$; that is, $(x_1, \ldots, x_s)$ is a dominating sequence of G and $s \leq \gamma_{\mathrm{gr}}(G)$. The same holds for side 1, so $|\sigma| \leq 2\gamma_{\mathrm{gr}}(G)$.

(ii) The vertices of $\mathsf{B}(G) \times \mathsf{B}(H)$ are the pairs $((g, a), (h, b))$, and two of them, $((g, a), (h, b))$ and $((g', a'), (h', b'))$, are adjacent exactly when $a' \neq a$, $b' \neq b$, $g' \in N[g]$ and $h' \in N[h]$. Hence $a + b \pmod 2$ is constant on components, and for $c \in \{0, 1\}$ the map $((g, a), (h, a + c)) \mapsto ((g, h), a)$ is an isomorphism of the part with $a + b \equiv c$ onto $\mathsf{B}(G \boxtimes H)$, since $N[g] \times N[h] = N_{G \boxtimes H}[(g, h)]$. The value of γ^t_{gr} follows from (i) and Lemma 7.1. □

Proof of Theorem D(i)*, total part.* Let $C > 0$ and let G, H satisfy $\gamma_{\mathrm{gr}}(G \boxtimes H) > C\gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H)$, as in Theorem 6.1. By Proposition 7.5,

$$\gamma^t_{\mathrm{gr}}(\mathsf{B}(G) \times \mathsf{B}(H)) = 4\gamma_{\mathrm{gr}}(G \boxtimes H) > 4C\gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H) = C\,\gamma^t_{\mathrm{gr}}(\mathsf{B}(G))\,\gamma^t_{\mathrm{gr}}(\mathsf{B}(H)).$$

The graphs $\mathsf{B}(G)$ and $\mathsf{B}(H)$ have no isolated vertices, and they are bipartite. □

Proposition 7.5 is a concrete form of the observation of [7] that Conjecture 1.2 for bipartite graphs is equivalent to Conjecture 1.1, and the proof of (i) is the bipartite splitting used there. It turns every counterexample to Conjecture 1.1 into one to Conjecture 1.2 with the same ratio. The smallest strong-product counterexample of Theorem 6.10, $K_{2,2,2} \boxtimes H_{11}$, gives $\mathsf{B}(K_{2,2,2}) \times \mathsf{B}(H_{11})$, a graph on $12 \cdot 22 = 264$ vertices with $\gamma^t_{\mathrm{gr}} \geq 44$, while $\gamma^t_{\mathrm{gr}}(\mathsf{B}(K_{2,2,2}))\,\gamma^t_{\mathrm{gr}}(\mathsf{B}(H_{11})) = 4 \cdot 10 = 40$. The authors of [7] checked Conjecture 1.2 by computer for all pairs of bipartite graphs with at most ten vertices, which is consistent with this size.

The order of magnitude transfers as well. The projection argument of Proposition 6.4 works for open neighbourhoods.

Proposition 7.6. *For all graphs G and H without isolated vertices,*

$$\gamma^t_{\mathrm{gr}}(G \times H) \le \min\{|V(G)|\,\gamma^t_{\mathrm{gr}}(H),\ \gamma^t_{\mathrm{gr}}(G)\,|V(H)|\},$$

so $\gamma^t_{\mathrm{gr}}(G \times H)/(\gamma^t_{\mathrm{gr}}(G)\gamma^t_{\mathrm{gr}}(H)) \le \frac{1}{2}\min\{|V(G)|, |V(H)|\}$.

Proof. Let $((g_1, h_1), \dots, (g_\ell, h_\ell))$ be a total dominating sequence of $G \times H$ with total footprints (a_i, b_i), and for $a \in V(G)$ let $I_a = \{i : a_i = a\}$. If $j < i$ lie in I_a, then $(a, b_i) \notin N(g_j) \times N(h_j)$ while $a = a_j \in N(g_j)$, so $b_i \notin N(h_j)$; and $b_i \in N(h_i)$. So $(h_i)_{i \in I_a}$ is a total dominating sequence of H and $|I_a| \le \gamma^t_{\mathrm{gr}}(H)$. Summing over a gives the first bound, and the second is symmetric. The ratio bound follows from $\gamma^t_{\mathrm{gr}}(G) \ge 2$, which holds since G has an edge. □

Corollary 7.7. *Let $\rho^t(n)$ be the largest value of $\gamma^t_{\mathrm{gr}}(G \times H)/(\gamma^t_{\mathrm{gr}}(G)\gamma^t_{\mathrm{gr}}(H))$ over graphs G, H without isolated vertices with $|V(G)||V(H)| \le n$. Then $\rho^t(n) \le \frac{1}{2}n^{1/2}$, and for every $\varepsilon > 0$ there is $c'_\varepsilon > 0$ with $\rho^t(n) \ge c'_\varepsilon n^{1/2-\varepsilon}$ for all $n \ge 16$, already for bipartite G and H.*

Proof. The upper bound is Proposition 7.6. For the lower bound let $N = \lfloor (n/4)^{1/2} \rfloor \ge 2$, take G, H with N vertices as in Theorem 6.3 for primes with $\theta_{p,q} > 1 - 2\varepsilon$, and pass to $\mathsf{B}(G)$ and $\mathsf{B}(H)$, which are bipartite with $2N$ vertices each, so $|V(\mathsf{B}(G))||V(\mathsf{B}(H))| = 4N^2 \le n$. By Proposition 7.5 the ratio for $\mathsf{B}(G) \times \mathsf{B}(H)$ equals $\gamma_{\mathrm{gr}}(G \boxtimes H)/(\gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H))$, which is at least $N^{\theta_{p,q}}/(pq+1)^{p+q-2}$, and $N \ge (n/4)^{1/2}/2$. □

In the hypergraph formulation of [7] the closed neighbourhoods of $G \boxtimes H$ are the products $N[g] \times N[h]$, so the same graphs give families of sets whose product family has the same ratio.

8. Projective planes

Let $q = p^e$ with p prime, let $\mathcal{P}$ and $\mathcal{L}$ be the sets of points and lines of $\mathrm{PG}(2, q)$, each of size $q^2 + q + 1$, and let N be the $\mathcal{P} \times \mathcal{L}$ incidence matrix, with $N_{x\ell} = 1$ if $x \in \ell$ and 0 otherwise. The incidence graph $\mathrm{Inc}(q)$ has vertex set $\mathcal{P} \cup \mathcal{L}$ and an edge $x\ell$ whenever $x \in \ell$. Every point is on $q+1$ lines and every line has $q+1$ points, two points are on exactly one line and two lines meet in exactly one point. So $\mathrm{Inc}(q)$ is $(q+1)$-regular and bipartite, it has no 4-cycles, and it has girth 6; distinct vertices have distinct open neighbourhoods.

8.1. **Grundy coverings.** In a bipartite graph the terms of a total dominating sequence on one side dominate only vertices of the other side. As in the proof of Proposition 7.5, γ^t_{gr} of a bipartite graph is therefore the sum of two independent quantities. For $\mathrm{Inc}(q)$, call a sequence of points $(x_1, \dots, x_s)$ a *Grundy covering* if for each i some line through x_i misses $x_1, \dots, x_{i-1}$, and let $c(q)$ be the largest length of a Grundy covering. Dually one defines Grundy coverings of points by lines, and since $\mathrm{PG}(2, q)$ is isomorphic to its dual their largest length is again $c(q)$. Hence

$$\gamma^t_{\mathrm{gr}}(\mathrm{Inc}(q)) = 2c(q). \tag{8.1}$$

Lemma 8.1. *For every field $\mathbb{F}$ and every matrix $M \in \mathbb{F}^{\mathcal{P}\times\mathcal{L}}$ whose nonzero entries are exactly the incident pairs, $c(q) \le \mathrm{rank}_{\mathbb{F}} M$.*

Proof. Let $(x_1, \dots, x_s)$ be a Grundy covering and let ℓ_i be a line through x_i missing $x_1, \dots, x_{i-1}$. The lines ℓ_i are distinct, since $x_j \in \ell_j$ and $x_j \notin \ell_i$ for $j < i$. The submatrix of M on rows $x_1, \dots, x_s$ and columns $\ell_1, \dots, \ell_s$ has nonzero diagonal and vanishes above the diagonal, because $M_{x_j \ell_i} = 0$ for $j < i$. □

Over a field whose characteristic does not divide q the incidence matrix itself is useless here: $NN^{\mathsf T} = qI + J$ has eigenvalues $(q+1)^2$ and q, so N is nonsingular. Over $\mathbb{F}_p$ its rank is small. Hamada's formula in this case is due to Smith [21] and, independently, to MacWilliams and Mann [20]:

$$\operatorname{rank}_{\mathbb{F}_p} N = \binom{p+1}{2}^{e} + 1. \tag{8.2}$$

For $e = 1$ this is $\binom{p+1}{2} + 1 = \frac{1}{2}(p^2 + p + 2)$, about half of $q^2 + q + 1$; for $p = 2$ it is $3^e + 1$.

Proof of Theorem E. By (8.1), Lemma 8.1 applied to N over $\mathbb{F}_p$, and (8.2),

$$\gamma^t_{\mathrm{gr}}(\mathrm{Inc}(q)) = 2c(q) \le 2\operatorname{rank}_{\mathbb{F}_p} N = 2\Big(\tbinom{p+1}{2}^{e} + 1\Big).$$

Since $|V(\mathrm{Inc}(q))| = 2(q^2+q+1) > 2p^{2e}$,

$$\frac{\gamma^t_{\mathrm{gr}}(\mathrm{Inc}(p^e))}{|V(\mathrm{Inc}(p^e))|} < \Big(\frac{p+1}{2p}\Big)^{e} + p^{-2e} \longrightarrow 0 \qquad (e \to \infty).$$

The values for $q \in \{2, 3, 4, 8\}$ are given in Proposition 8.2. □

The bound $c(q) \le \operatorname{rank}_{\mathbb{F}} M$ holds for every matrix with the incidence pattern, not only for N. Over $\mathbb{F}_p$ one cannot do better with another pattern matrix in the cases where the bound is attained, and in those cases the rank of the incidence matrix is the minimum rank of the pattern. For $q = 5$ it is not attained.

Proposition 8.2.

q	2	3	4	5	8
$\lvert V(\mathrm{Inc}(q))\rvert$	14	26	42	62	146
$\operatorname{rank}_{\mathbb{F}_p} N$	4	7	10	16	28
$c(q)$	4	7	10	15	28
$\gamma^t_{\mathrm{gr}}(\mathrm{Inc}(q))$	8	14	20	30	56

Proof. The upper bounds for $q \neq 5$ are (8.2). The values of $c(q)$ for $q \le 5$ were computed by an exhaustive search over the sets of covered lines (Appendix C), which visits 704 states for $q = 3$ and about 1.6×10^6 for $q = 5$ and also confirms the upper bound $c(5) \le 15$. For $q = 8$ a Grundy covering of length 28 is listed in Appendix A; it was found by a randomised greedy search and is checked directly, and (8.2) shows that it is longest. □

For $q = 2$ the graph $\mathrm{Inc}(2)$ is the Heawood graph. Let m be a line of the Fano plane, x_1, x_2, x_4 its points and x_3 a point off m. Then (x_1, x_2, x_3, x_4) is a Grundy covering of length 4 (Figure 8). Indeed x_1 is on three lines; of the three lines through x_2 only m contains x_1; of those through x_3 only x_3x_1 and x_3x_2 meet $\{x_1, x_2\}$; and of those through x_4, the line m contains x_1 and x_2, the line x_4x_3 contains x_3, and the third line misses all three.

Remark 8.3. Theorem E answers the question of [11] for bipartite graphs of girth 6 and minimum degree $q + 1$, with q arbitrarily large. The graphs are regular, have no open twins and are vertex-transitive, so the obstruction is not a local one. For $p = 2$ the density $\gamma^t_{\mathrm{gr}}/|V|$ is at most $(3/4)^e$ up to a negligible error. The exact values in Proposition 8.2 suggest that $c(2^e) = 3^e + 1$ for all e (Problem 10.4).

9. Computations

None of the proofs depends on a computer. The computations below were used to find the constructions and to test the statements, and they are reproducible from the repository named in the data availability statement.

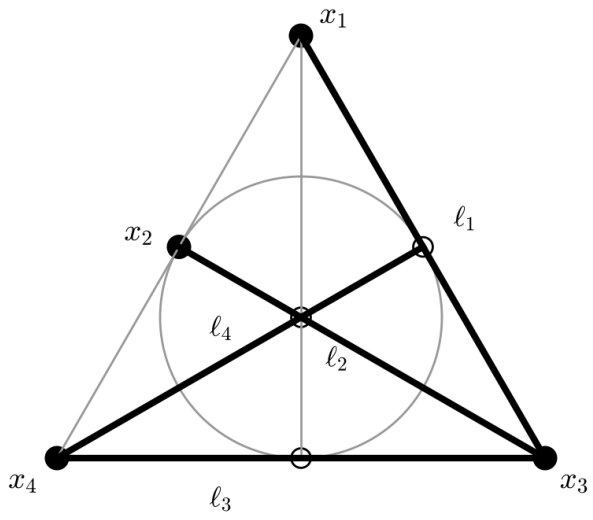


FIGURE 8. A Grundy covering of length $4 = \operatorname{rank}_{\mathbb{F}_2} N$ in the Fano plane. The points x_1, x_2, x_4 lie on a line m and x_3 lies off it; the thick lines $\ell_1, \dots, \ell_4$ satisfy $x_i \in \ell_i$ and $\ell_i \cap \{x_1, \dots, x_{i-1}\} = \emptyset$.

TABLE 5. The octahedron times H_N. The row $2\gamma_{\mathrm{gr}}(H_N)$ is the value predicted by Conjecture 1.1, and the last row the exact value, which equals $\max(N, 2\gamma_{\mathrm{gr}}(H_N))$ in this range; bold entries are counterexamples.

N	4	5	6	7	8	9	10	11	12	13	14	15
$\gamma_{\mathrm{gr}}(H_N)$	3	3	4	4	4	5	5	5	6	6	6	7
$2\gamma_{\mathrm{gr}}(H_N)$	6	6	8	8	8	10	10	10	12	12	12	14
$\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N)$	6	6	8	8	8	10	10	**11**	12	**13**	**14**	**15**

9.1. **An exact solver.** A dominating sequence is determined, as far as its continuations are concerned, by the set D of vertices it has dominated so far: a vertex v can be appended exactly when $N[v] \not\subseteq D$, and the new set is $D \cup N[v]$. So $\gamma_{\mathrm{gr}}(G) = f(\emptyset)$ for the function on subsets

$$f(D) = \max\bigl(\{0\} \cup \{1 + f(D \cup N[v]) : N[v] \not\subseteq D\}\bigr),$$

and the same recursion with open neighbourhoods gives γ^t_{gr}. We evaluate f by depth-first search with a hash table keyed by D, stored as a 128-bit mask, so graphs with up to 128 vertices can be handled; the cost is the number of reachable sets D. Two simple reductions make the search feasible for the graphs of this paper: closed twins are merged (Remark 2.6), and for bipartite graphs and γ^t_{gr} the two sides are treated separately, which is (8.1). The values obtained are:

- $\gamma_{\mathrm{gr}}(C_k \boxtimes C_\ell) = (k-2)(\ell-2)$ for $3 \le k \le \ell \le 7$, and $\gamma_{\mathrm{gr}}(P \boxtimes C_5) = 15$, where P is the Petersen graph and $\gamma_{\mathrm{gr}}(P) = 5$;
- γ^t_{gr} of the direct products in Table 4, $\gamma^t_{\mathrm{gr}}(C_n) = 2\lfloor (n-1)/2 \rfloor$ for $3 \le n \le 14$, and $\gamma^t_{\mathrm{gr}}(T) = 2\nu(T)$ for thirty random trees with up to sixteen vertices;
- $\gamma_{\mathrm{gr}}(H_N) = \lfloor N/3 \rfloor + 2$ for $4 \le N \le 24$, and the values of $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N)$ in Table 5;
- $c(q)$ for $q \in \{2, 3, 4, 5\}$ in Proposition 8.2.

9.2. **Ranks.** The ranks used in Sections 3, 6 and 8 were recomputed independently: by Gaussian elimination over the prime fields for $A_0 + I$ of $K(n,2)$ over $\mathbb{F}_2$ ($n = 6, 8, \dots, 14$), for $A_0 - I$ over $\mathbb{F}_p$ for the pairs $(n,p) = (7,3), (9,5), (10,3), (11,7), (13,3), (14,5)$, for the matrices of Theorem 6.1 with $7 \le m \le 10$, and for the p-ranks (8.2) with $q \in \{2,3,4,5,7,8\}$; and by computing eigenvalues for the circulants $A - 2\cos(2\pi/k)I$ of C_k and the matrices A_ω of Proposition 7.3, for $3 \le k, n \le 30$. The factorizations behind the rank bounds for $K(n,2)$ with $n \in \{6,7,8,9,10\}$ and for the Fano plane are certified in Lean (Appendix C).

9.3. **Kneser graphs.** The ranks of the matrices $I + cD_{n,r}$ in Section 4 were computed by Gaussian elimination over $\mathbb{F}_p$ in two independent implementations, one using the FLINT library and one written in Julia (Appendix C). The dominating sequences of Table 2 were found by a randomised greedy procedure that starts from a random dominating prefix and then repeatedly appends a vertex with the fewest newly dominated vertices; the five sequences of Theorem 4.9 are listed in Appendix B. For the matrices of Theorem 6.1 the ranks over $\mathbb{F}_2$ and $\mathbb{F}_3$ are exactly m and $\binom{m}{2}$ for $8 \le m \le 14$.

9.4. **A search for small counterexamples.** Let $\mathcal{G}$ be the set of connected graphs with 4 to 7 vertices that are not forests and have no closed twins; forests are excluded because Conjecture 1.1 holds when one factor is a forest [3], and closed twins because they do not change the problem (Remark 2.6). There are 560 such graphs. For each of the 157,080 unordered pairs $\{G,H\} \subseteq \mathcal{G}$ we computed $\gamma_{\mathrm{gr}}(G)$ and $\gamma_{\mathrm{gr}}(H)$ exactly and ran a randomised greedy search for a dominating sequence of $G \boxtimes H$ longer than $\gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H)$. None was found. This is not a proof that the conjecture holds for these pairs, but together with Theorem B it suggests that counterexamples need a factor with a large gap between γ_{gr} and mr, as H_N has, and are not small. The smallest counterexample we know, $K_{2,2,2} \boxtimes H_{11}$, has 66 vertices.

10. Status of the conjectures and open problems

Table 6 records the status of every conjecture and question on Grundy domination of products and of bipartite graphs treated in this paper.

Problem 10.1. *Find the smallest number of vertices of a strong product $G \boxtimes H$ with $\gamma_{\mathrm{gr}}(G \boxtimes H) > \gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H)$. We know that it is at most 66, and Section 9 suggests that it exceeds 49.*

Problem 10.2. *Characterise the graphs that are $\mathbb{F}$-tight for some field $\mathbb{F}$. In particular, is every tree tight over some field? A positive answer would give a new proof of the theorem of Bell, Driscoll, Krop and Wolff [3] restricted to tight second factors. Paths and stars are tight over $\mathbb{R}$ (Propositions 3.3 and 3.4).*

Problem 10.3. *Can the loss $n^{-\varepsilon}$ in Corollary 6.5 be removed, that is, is there $c > 0$ with $\rho(n) \ge cn^{1/2}$ for all n? Corollary 6.5 gives $c_\varepsilon n^{1/2-\varepsilon} \le \rho(n) \le n^{1/2}$.*

Problem 10.4. *Is $c(2^e) = 3^e + 1$ for every e, so that $\gamma^t_{\mathrm{gr}}(\mathrm{Inc}(2^e)) = 2(3^e+1)$? What is $c(p)$ for odd primes p? We have $c(5) = 15$, one less than the 5-rank, and for $q = 7$ a randomised search found a Grundy covering of length 25, while the 7-rank is 29.*

Problem 10.5. *For a graph G let $g(G) = \min_{\mathbb{F}} \mathrm{mr}_{\mathbb{F}}(G) - \gamma_{\mathrm{gr}}(G)$, the minimum over all fields. Corollary 6.11 gives $g(H_N) \ge N/2 - \lfloor N/3 \rfloor - 2$. Is there an $\epsilon > 0$ and graphs of every order with $g(G) \ge \epsilon |V(G)|$? Large gaps are what counterexamples to Conjecture 1.1 are made of, by Proposition 5.5.*

Problem 10.6. *Is $\gamma_{\mathrm{gr}}(K(14,4))$ equal to 286 or to 287? This is the first open case of the conjecture of Campêlo and Severín. A dominating sequence of length 287 would disprove it; if*

TABLE 6. Conjectures and questions treated in this paper.

statement	status	where
$\gamma_{\mathrm{gr}}(G \boxtimes H) = \gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H)$ for all G, H [8, Conj. 22]	false; largest ratio $n^{1/2+o(1)}$	Theorems 6.1, 6.10, Corollary 6.5
$\gamma_{\mathrm{gr}}(C_k \boxtimes C_\ell) = (k-2)(\ell-2)$ (conjectural value in [8])	true	Theorem B
$\gamma^t_{\mathrm{gr}}(G \times H) = \gamma^t_{\mathrm{gr}}(G)\gamma^t_{\mathrm{gr}}(H)$ [7, Conj. 2.2]	false; largest ratio $n^{1/2+o(1)}$	Proposition 7.5, Corollary 7.7
$\gamma^t_{\mathrm{gr}}(C_k \times C_\ell) = \gamma^t_{\mathrm{gr}}(C_k)\gamma^t_{\mathrm{gr}}(C_\ell)$ [7]	extended to any number of cycles, forests, complete and complete bipartite graphs	Theorem 7.4
Is there $c > 0$ with $\gamma^t_{\mathrm{gr}}(G) \geq c\lvert V(G)\rvert$ for bipartite C_4-free G with $\delta(G) \geq 2$? [11, Question 1]	no	Theorem E
$\gamma_{\mathrm{gr}}(K(n,2)) = n-1$ for $n \geq 6$ [13]	new proof of the upper bound	Proposition 3.8
$\gamma_{\mathrm{gr}}(K(n,r)) = \binom{n-1}{r-1}$ for $n \geq 3r$ [14, Sect. 6.5]	true unless $n-2r$ divides $\mathrm{lcm}(1,\dots,r-1)$; open for $(14,4)$	Theorem 4.4, Corollary 4.5
threshold $\bar{n}_r$ for $\gamma_{\mathrm{gr}}(K(n,r)) = \alpha$ [13]	$\bar{n}_r \leq 2r + \mathrm{lcm}(1,\dots,r-1) + 1$	Corollary 4.5
optimality of the tabu sequences for $K(8,3)$, $K(9,4)$, $K(10,4)$, $K(11,4)$, $K(11,5)$ [14]	all optimal	Theorem 4.9
dependence of $\gamma^Z_{\mathrm{gr}}(K(n,r))$ on n for $2r+1 \leq n \leq 3r$ [13, Sect. 7]	for $r=2$: constant 5, below $\binom{2r}{r} = 6$	Proposition 4.12

there is none, a certificate other than a rank bound is needed, since the least rank we found among the matrices $I + cD_{14,4}$ is 287.

Problem 10.7. *Is $\gamma_{\mathrm{gr}}(K(n,r)) = \min_{p,c} \mathrm{rank}_{\mathbb{F}_p}(I + cD_{n,r})$ for all $n \geq 2r+1$, the minimum over primes p and $c \in \mathbb{F}_p^\times$? Equality holds in every case where both sides are known (Theorem 4.4 and Table 2).*

APPENDIX A. EXPLICIT SEQUENCES

The Petersen graph. With the vertices of the Petersen graph written as the two-element subsets of $\{1,\dots,5\}$, adjacent when disjoint, the sequence $\{1,2\},\{1,3\},\{2,4\},\{1,5\},\{1,4\}$ is a dominating sequence with footprints $\{1,2\},\{1,3\},\{1,5\},\{2,3\},\{1,4\}$, so $\gamma_{\mathrm{gr}} \geq 5$; Proposition 3.8 gives equality.

The octahedron times H_{11}. In the notation of Section 6, the eleven vertices $(g_{-j \bmod 3}, j)$, $0 \leq j \leq 10$, in increasing order of j, form a dominating sequence of $K_{2,2,2} \boxtimes H_{11}$. Footprints are $(g_0,0)$, $(g_0,2)$, $(g_0,3)$ for $j = 0,1,2$ and (x_{1-j}, j) for $j \geq 3$; the reader can check this from (6.2) and the fact that x_t is adjacent to every vertex of the octahedron except g_t. The sequence is certified in Lean.

A Grundy covering of PG$(2,8)$**.** Write $\mathbb{F}_8 = \mathbb{F}_2[x]/(x^3+x+1)$ and represent $a_0+a_1x+a_2x^2$ by the integer $a_0+2a_1+4a_2$. A point (p_0,p_1,p_2) lies on the line $[l_0,l_1,l_2]$ when $p_0l_0+p_1l_1+p_2l_2=0$. Table 7 lists a Grundy covering $(x_1,\dots,x_{28})$ of length 28 together with lines $\ell_i \ni x_i$ that miss $x_1,\dots,x_{i-1}$.

TABLE 7. A Grundy covering of PG$(2,8)$ of length $28 = 3^3+1$.

i	x_i	ℓ_i	i	x_i	ℓ_i
1	$(1,2,3)$	$[0,1,7]$	2	$(1,0,1)$	$[0,1,0]$
3	$(1,5,7)$	$[0,1,2]$	4	$(1,3,7)$	$[1,1,3]$
5	$(1,5,1)$	$[0,1,5]$	6	$(1,0,2)$	$[1,0,5]$
7	$(1,4,3)$	$[1,2,7]$	8	$(1,4,4)$	$[0,1,1]$
9	$(1,3,4)$	$[1,7,2]$	10	$(1,3,0)$	$[0,0,1]$
11	$(1,6,7)$	$[1,3,0]$	12	$(1,5,5)$	$[1,0,2]$
13	$(1,2,0)$	$[1,5,4]$	14	$(1,6,3)$	$[1,2,2]$
15	$(1,1,5)$	$[1,1,0]$	16	$(1,1,2)$	$[1,2,4]$
17	$(1,1,1)$	$[1,7,6]$	18	$(1,3,5)$	$[0,1,6]$
19	$(1,0,7)$	$[1,7,4]$	20	$(1,2,6)$	$[1,0,3]$
21	$(1,7,1)$	$[1,4,0]$	22	$(1,7,5)$	$[1,1,7]$
23	$(1,3,6)$	$[1,3,7]$	24	$(1,2,1)$	$[1,4,2]$
25	$(1,0,5)$	$[1,1,2]$	26	$(0,0,1)$	$[1,0,0]$
27	$(1,7,3)$	$[0,1,4]$	28	$(0,1,6)$	$[0,1,3]$

APPENDIX B. DOMINATING SEQUENCES OF KNESER GRAPHS

The ground set is $\{1,\dots,9,a,b\}$ with $a=10$, $b=11$, and an r-set is written as the string of its elements. Each entry S/F is a term S of the sequence followed by a footprint F, a vertex of $N[S]$ (that is, $F=S$ or $F\cap S=\emptyset$) lying in no closed neighbourhood of an earlier term. Terms are listed in order, row by row.

$K(8,3)$**, length** 22**.** 136/136 123/123 135/135 134/134 138/138 137/137 467/125 347/347 457/236 478/356
147/147 125/346 127/127 126/357 124/124 568/234 456/178 345/167 358/146 246/157 378/145 257/148

$K(9,4)$**, length** 77**.** 1248/1248 1347/1347 3579/1246 1246/3578 1247/1247 1249/1249 1245/1245 1234/1234
5689/1237 3678/1259 6789/1235 3689/1257 2689/1357 4689/4689 1689/1689 1357/2469 3457/1269 3567/1489
2357/1469 5789/1236 2469/1358 1378/2456 3789/1256 1237/4568 2378/2378 3478/3478 4569/1238 2569/1348
3569/2478 1569/2348 1379/1379 1367/1367 2458/1369 1369/2578 1269/3458 1469/2358 2589/1467 2369/2369
2679/2679 3679/1258 4578/1239 3469/3469 3467/1589 4679/4679 2367/2367 2347/1568 3479/3479 1589/2467
3589/1267 2358/1479 2578/1349 1578/2349 1358/2479 1568/2379 4568/1279 1256/3489 1267/3459 1268/4579
1238/4567 1257/3468 1289/3456 1235/4678 1349/2567 1459/2368 1348/2579 1489/2356 4579/1368 3459/1678
1457/2389 1456/2789 2457/1389 2789/1356 2489/1567 3456/1789 2348/1579 2467/1359 2359/2359

$K(10,4)$**, length** 93**.** 3458/3458 1489/1489 1346/1346 1348/1348 1248/1248 1478/1478 1458/1458 148a/148a
1468/1468 1234/1234 1345/1345 134a/134a 1349/1349 1347/1347 3459/3459 2345/2345 345a/345a 3457/3457
3456/3456 1789/2346 178a/2349 1678/249a 1278/469a 1378/1378 1578/1578 168a/2347 1268/347a 1689/247a
1368/1368 1568/1568 189a/2467 128a/4679 1289/467a 1358/1358 1258/1258 158a/158a 138a/138a 1589/1589
1238/1238 1389/1389 346a/1257 3467/125a 3469/157a 3468/3468 349a/1256 2349/156a 347a/1569 3489/3489
348a/348a 3478/3478 2348/2348 679a/1235 579a/1236 379a/1456 789a/1356 279a/3568 3679/124a 1679/238a
5679/123a 4679/135a 2479/136a 5789/146a 2379/468a 1479/1479 1379/1379 3789/3789 4789/4789 578a/1239
278a/1359 378a/378a 478a/478a 4578/4578 4678/4678 1359/2478 2678/149a 2378/2378 1259/3678 1469/3578
236a/1457 238a/1467 2369/147a 268a/1357 367a/2489 2469/137a 356a/1247 259a/1367 569a/1237 567a/2389
146a/3589 147a/3689 1456/389a 2356/489a 235a/4689

$K(11,4)$, **length** 121. 1489/1489 578b/578b 238a/238a 4789/4789 2489/2489 4589/4589 3489/3489 489b/489b
4689/4689 489a/489a 248a/248a 478b/478b 478a/478a 468a/468a 348a/348a 48ab/48ab 458a/458a 148a/148a
468b/468b 348b/348b 458b/458b 148b/148b 248b/248b 2468/2468 2348/2348 2458/2458 1248/1248 2478/2478
4578/4578 1468/1468 1478/1478 3468/3468 3478/3478 1348/1348 3458/3458 4568/4568 4678/4678 1458/1458
378b/378b 378a/378a 3578/3578 1378/1378 3678/3678 3789/3789 2378/2378 2578/2578 1578/1578 578a/578a
5678/5678 5789/5789 258a/258a 1258/1258 2358/2358 2589/2589 2568/2568 258b/258b 238b/238b 2368/2368
2389/2389 1238/1238 138b/138b 138a/138a 1358/1358 1368/1368 1389/1389 158a/158a 1568/1568 158b/158b
1589/1589 5689/5689 58ab/58ab 389a/389a 38ab/38ab 589b/589b 3568/3568 389b/389b 568a/568a 368a/368a
589a/589a 358a/358a 3689/3689 358b/358b 568b/568b 368b/368b 3589/3589 267b/1345 26ab/3457 126b/789a
269b/178a 268b/268b 1268/1268 268a/268a 269a/178b 2679/18ab 267a/189b 2367/89ab 246a/789b 2356/78ab
2689/2689 2678/2678 149a/278b 278b/3456 12ab/6789 2789/2789 27ab/1689 146a/289b 23ab/1678 179a/2345
128b/128b 1289/1289 145a/689b 369a/1278 257a/168b 1279/68ab 129b/678a 127b/689a 289a/289a 1457/28ab
146b/278a 2357/168a 4579/128a

$K(11,5)$, **length** 296. 1278a/1278a 3569b/12478 4569b/12378 3469b/12578 12378/4569a 12478/3569a 12578/3469a
12789/12789 12678/12678 1278b/1278b 3459a/1267b 569ab/12347 3456b/1279a 349ab/12567 346ab/12579
469ab/12357 1346b/1346b 3467b/3467b 3468b/3468b 2346b/2346b 345ab/12679 1589a/23467 5679b/5679b
5689b/5689b 2569b/2569b 1569b/1569b 1347a/25689 134ab/134ab 1348a/2567b 1345a/1345a 1234a/5678b
1349a/1349a 1346a/1346a 25789/136ab 347ab/347ab 348ab/348ab 234ab/234ab 15689/237ab 2678b/13459
56789/1234b 25689/1347b 25678/1349b 35689/35689 45689/45689 5689a/5689a 3569a/1248b 1237b/4568a
4589a/12367 5789a/12346 2589a/13467 589ab/589ab 3589a/3589a 1247b/3568a 12347/568ab 23578/1469a
2578a/13469 2578b/1369a 24578/24578 3456a/1279b 12346/578ab 12357/4689b 12367/4589b 1237a/4568b
12379/12379 12579/3468a 1279b/34568 1279a/3458b 12679/358ab 12479/12479 3458a/1269b 1247a/3589b
12467/389ab 12457/12457 3457a/3457a 2345a/2345a 1267b/34589 127ab/34689 12567/3489b 1257b/1257b
1267a/1267a 3469a/1258b 3468a/1259b 2348a/1567b 3478a/1256b 2349a/2349a 2347a/1589b 3479a/3479a
2346a/2346a 3467a/3467a 5789b/1236a 1589b/2367a 4589b/1367a 3589b/1467a 12589/367ab 24589/24589
23589/23589 369ab/12458 15789/236ab 45789/45789 35789/35789 34589/267ab 13589/13589 14589/14589
2367b/1458a 2679b/13458 2467b/1389a 2567b/13489 146ab/23579 236ab/14579 246ab/13579 456ab/12389
467ab/12359 136ab/24579 356ab/12489 367ab/12459 1258b/34679 568ab/12349 567ab/567ab 12348/5679a
156ab/156ab 256ab/256ab 12358/4679b 126ab/34578 1234b/5678a 12349/678ab 12345/12345 678ab/23459
679ab/23458 169ab/23457 168ab/168ab 689ab/13457 268ab/268ab 269ab/269ab 34789/1256a 13478/2569a
13458/2679a 2568b/1379a 2356b/2356b 1256b/3789a 2456b/2456b 3458b/1269a 34578/129ab 34568/179ab
2458b/2458b 23478/1569a 3478b/3478b 34678/34678 5678b/1239a 3568b/2479a 4568b/2379a 23489/1567a
2348b/2348b 1569a/2378b 3489b/2567a 34689/257ab 1348b/1348b 23468/23468 2569a/1378b 13468/13468
5679a/1238b 23479/1568a 23467/189ab 23457/1689a 579ab/12368 6789b/1235a 4679b/2358a 1679b/2458a
3679b/3679b 2458a/1379b 12458/379ab 1258a/3479b 24568/24568 1379a/2468b 1579a/2368b 12569/378ab
1349b/2568a 13479/258ab 3479b/1268a 13467/289ab 3459b/1678a 2568a/1479b 34579/128ab 13469/278ab
258ab/14679 13457/2689a 34567/34567 23459/178ab 3457b/3457b 23568/23568 23456/23456 2358b/2358b
23469/23469 2349b/2349b 1345b/1345b 1679a/2345b 13456/13456 2679a/1358b 4679a/1235b 3679a/1245b
189ab/23567 1389a/2457b 1369a/4578b 1689a/2357b 4689a/1357b 1289a/3567b 1269a/3578b 2689a/1457b
2469a/2469a 1578b/2369a 3578b/12469 358ab/24679 578ab/12369 3567b/2489a 458ab/23679 1367b/2459a
1567b/2389a 3678b/3678b 1259a/4678b 12459/3678a 1239a/45678 1678b/2359a 1359a/24678 359ab/14678
1356b/1356b 2489a/13567 1245b/36789 125ab/46789 12359/4678a 1249a/35678 1456b/1456b 1235a/4789b
4789a/12356 3789a/12456 1256a/3789b 2359a/1468b 124ab/35679 389ab/14567 1459a/23678 1236a/4579b
2379a/14568 3579a/3579a 1248a/3579b 128ab/45679 12468/357ab 459ab/13678 479ab/13568 12689/457ab
2459a/1368b 4579a/4579a 249ab/249ab 15678/239ab 45678/1239b 14578/2369b 12389/4567a 35678/1249b
2368b/1457a 1289b/3567a 4568a/2379b 2689b/1357a 4578a/4578a 1468b/2357a 3689b/2457a 13578/2469b
3568a/2479b 1249b/3578a 1357a/24689 2489b/1356a 135ab/135ab 1357b/2468a 1579b/2368a 3567a/1489b
2567a/1389b 24567/138ab 1389b/2456a 2356a/2356a 1459b/1459b 24678/1359b 4567a/2389b 24579/1368a
26789/145ab 138ab/24569 378ab/14569 2457a/13689 23689/23689 13579/248ab 13689/245ab 178ab/23569
15679/238ab 1456a/1456a 23567/148ab 16789/235ab 14678/2359b 1468a/1468a 3678a/2459b 2478a/13569
2357a/14689

Appendix C. Machine checks

Three independent programs were used; none of the proofs relies on them.

The exact solver. A C program evaluates the recursion of Section 9 with 128-bit masks and a hash table. A Python driver produces the graphs and runs the randomised greedy search of Section 9.

Julia. A separate Julia program, written without reference to the C code, recomputes with its own memoised recursion the values of γ_{gr} for $C_k \boxtimes C_\ell$ $(3 \le k \le \ell \le 6)$, the Petersen graph and

its product with C_5, H_N $(4 \le N \le 20)$, $K_{2,2,2} \boxtimes H_N$ $(4 \le N \le 12)$ and $K(n,2)$ $(6 \le n \le 12)$; the values of γ_{gr}^t for C_n $(3 \le n \le 14)$ and for six direct products of cycles; the support patterns and ranks of the matrices of Theorem 6.1 for $7 \le m \le 10$; the ranks of Section 9; and $c(q)$ for $q \in \{2,3,5\}$. All 153 checks pass.

Lean. A Lean 4 file using only the core library certifies, by kernel evaluation (`decide`), the dominating sequences above, a dominating sequence of length 5 of H_{11}, the Grundy coverings of the Fano plane and of $\mathrm{PG}(2,8)$, and the factorizations $M = CR$ over $\mathbb{F}_p$, with C having $n-1$ columns, for $A_0 + I$ of $K(n,2)$ over $\mathbb{F}_2$ $(n = 6, 8, 10)$ and $A_0 - I$ over $\mathbb{F}_3$ $(n = 7, 10)$ and $\mathbb{F}_5$ $(n = 9)$, and with 4 columns for the incidence matrix of the Fano plane. The file has 13 theorems; twelve use no axioms and one uses only `propext`.

Kneser graphs. A second Lean 4 file, again using only the core library, encodes an r-subset of $\{0, \dots, n-1\}$ as a bit mask and certifies by kernel evaluation (`decide +kernel`) that the five sequences of Appendix B are dominating sequences of $K(8,3)$, $K(9,4)$, $K(10,4)$, $K(11,4)$ and $K(11,5)$, by scanning all vertices for footprints; its five theorems use only `propext`. A second Julia program recomputes by its own Gaussian elimination the rank $\binom{n-1}{r-1}$ of $M_{n,r}$ for $(n,r,p) = (9,3,3)$, $(10,3,2)$, $(11,3,5)$, $(12,4,2)$, $(13,4,5)$, $(16,4,2)$, $(15,5,5)$, every rank of Table 2 except the one for $K(16,5)$, the five sequences, and the ranks of the matrices of Theorem 6.1 for $8 \le m \le 14$ together with the absence of common edges of A_m and B_m. All its checks pass. The rank 1380 for $K(16,5)$, and all the other ranks, were also computed with FLINT, and the Z-Grundy numbers of Proposition 4.12 with the exact solver.

Author contributions

Deep Bhattacharjee is the sole author. He posed the problems in the form treated here, proved the results, wrote the paper, drew the figures, and wrote the programs and the Lean certificates.

Open Gluon and Anaxia

Two tools built by the author organised this project. Open Gluon is a workspace that keeps pinned copies of the mathematical libraries, the source record of the project and its checks in one reproducible place. Anaxia, the engine of the eBLLM project, stores each construction as a route with its input, its output, the obligations that must be discharged and a test that makes the route fail when an obligation is skipped, and it certifies finite statements in exact integer arithmetic. It was used here to keep the sequences, matrices and their checks in one record. Every statement of this paper is proved in the text.

Declarations

Data and code availability. No data were generated. The exact solver, the Julia programs and the Lean 4 certificates are available at `https://github.com/creelie/Paper_1` and archived on Zenodo under the DOI 10.5281/zenodo.23607651.

Funding. No funding of any kind was received for this work.

Competing interests. The author declares no competing interests.

Ethics approval. Not applicable. The work involves no human participants, animals or personal data.

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